BVAR Scenario Tool
User Manual and Methodological Documentation[1]

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Source code: `https://github.com/matyasfarkas/BVARScenarioTool`

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Abstract

This document describes the BVAR Scenario Tool, a browser-based application for conditional forecasting, scenario analysis, and policy counterfactual evaluation. The tool combines a large Bayesian vector autoregression (BVAR) estimated with hierarchical Minnesota priors (Giannone et al., 2015) with structural impulse responses drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). Users condition the BVAR forecast on arbitrary variable paths via a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012), and evaluate monetary and fiscal policy counterfactuals using DSGE-identified impulse response functions. We describe the methodology employed, the implementation of the conditional forecasting algorithms, and the three-layer architecture that separates the unconditional baseline, the user's scenario, and the policy counterfactual. All computation runs client-side in the browser; the tool requires no server infrastructure and can be deployed as a static website.

Keywords: BVARs, Conditional Forecasting, Scenario Analysis, Kalman Filter, Durbin–Koopman Smoother, Policy Counterfactuals, DSGE Models, MMB

JEL classification: C32, C53, E37, E52, E58

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Contents

1 Introduction

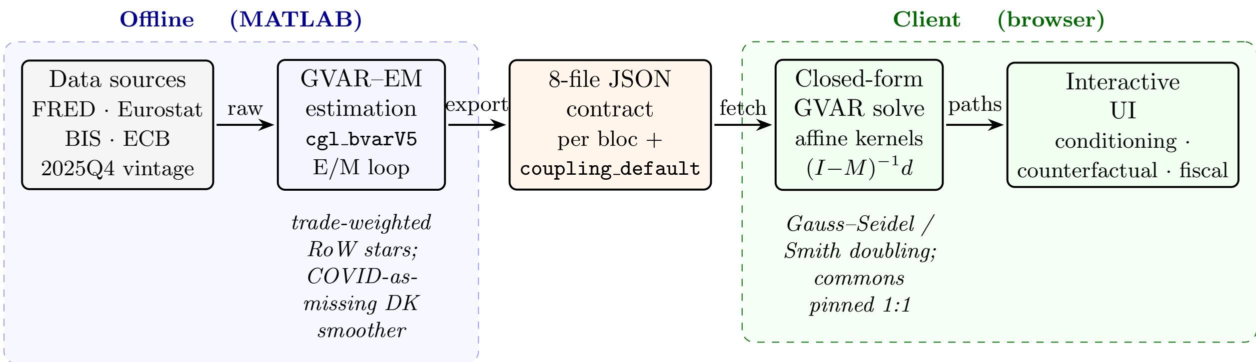

Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Harmonized macro-financial data (FRED, Eurostat, BIS, ECB; common 2025Q4 vintage) feed the *offline* MATLAB GVAR–EM estimator: per-bloc Bayesian VARs (`cgl_bvarV5`) are refit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world "star" variables (ROWDEM/ROWINF/ROWFIN) from partners' Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to $\max|\Delta\text{star}| < 0.03$ (cf. §6). Each converged bloc is serialized to the eight-file JSON contract (**`spec`**, **`companion_mean`**, **`companion_draws`**, **`baseline`**, **`baseline_levels`**, **`historical`**, **`historical_levels`**, **`transformations`**; plus **`fiscal_meta`** and the precomputed **`coupling_default.json.gz`** coupling artifact). In the *client* (browser), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§7).

Conditional forecasting with Bayesian vector autoregressions (BVARs) has become a standard tool in central bank forecasting and policy analysis. The seminal contributions of Doan et al. (1984) and Waggoner and Zha (1999) established the methodological foundations, while Bańbura et al. (2015) extended these techniques to large cross-sections using state-space methods. The BVAR Scenario Tool described in this document builds on this literature to provide an interactive, browser-based platform that makes conditional forecasting and policy counterfactual analysis accessible without specialized software.

The tool implements a three-layer analytical framework:

(i) **Baseline forecast:** An unconditional forecast from a large BVAR estimated on quarterly macroeconomic and financial data, following the hierarchical prior framework of Giannone et al. (2015).

(ii) **Conditional scenario:** The user conditions any subset of variables on desired paths; a Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) propagates these constraints to all remaining variables, respecting the full covariance structure of the VAR.

(iii) **Policy counterfactual:** An alternative monetary or fiscal policy path is overlaid on the scenario using impulse response functions from a DSGE model drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and the resulting macroeconomic responses are re-conditioned through the BVAR to ensure cross-variable consistency.

The tool is designed for economists at central banks and international institutions who need to construct and communicate conditional projections and policy scenarios. It runs entirely in the browser, requiring no server-side computation, and can be deployed as a static website on any hosting platform.

This document is organized as follows. Section 2 describes the BVAR estimation methodology. Section 3 derives the companion-form state-space representation. Section 4 presents the Kalman filter and Durbin–Koopman smoother used for conditional forecasting. Section 5 details the policy counterfactual framework. Section 9 documents the variable coverage. Section 10 provides a guide to the user interface. Section 11 discusses technical implementation details. Section 14 concludes with a discussion of limitations and directions for future development.

1.1 Notation

Table 1 collects the symbols used throughout the manual, grouped by the layer in which they appear (BVAR estimation, the companion-form state space and smoother, the GVAR coupling, the fiscal recursion, and the long-run tilt). A plain-language *Glossary* of the named concepts—BVAR, GVAR, the trade-weighted stars, COVID-as-missing, the stock-flow adjustment, the EM loop, the endogenous dominant unit, the entropic tilt, and the evaluation statistics (CRPS, Giacomini–White, . . .)—is provided in Appendix A.

Table 1: Notation used across the BVAR, GVAR, fiscal, and tilt layers.

Symbol	**Meaning**
BVAR estimation (Sec. 2)	
y_t	$n \times 1$ vector of endogenous variables of a bloc at quarter t (in estimation space).
n	number of variables in a bloc ($n = 41$ for the endogenous-US bloc).
p	VAR lag order ($p = 3$ throughout).
B_ℓ	$n \times n$ reduced-form autoregressive coefficient matrix on lag ℓ.
ε_t	reduced-form innovation, $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$.
Σ	$n \times n$ innovation covariance; posterior-mean $\hat{\Sigma}$.
β, $\hat{\beta}$	stacked VAR coefficients; posterior mean.
λ	overall Minnesota/GLP shrinkage (tightness) hyperparameter.
μ	sum-of-coefficients prior hyperparameter (Giannone et al., 2015).
ψ	residual-variance scale of the prior (per variable).
δ_i	prior mean on variable i's own first lag: $\delta_i = 1$ (random walk, `RW`) or $\delta_i = 0$ (stationary white noise, `WN`).
`pos`	index set of the `WN` variables; sets `diagb(pos)` $= 0$ (Sec. 2.5).

continued on next page

Table 1 continued from previous page

Symbol	Meaning
D	number of retained posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$.
State space & smoother (Sec. 3–4)	
α_t	companion-form state vector (stacked current and lagged y).
T, Z, Q, H	transition, observation, state-noise, and measurement-noise matrices.
$Z_t^{(o)}$	rows of Z corresponding to the *observed* variables at t (partial observation).
$\alpha_{t\|t-1}, P_{t\|t-1}$	one-step predicted state mean and covariance.
v_t, F_t, K_t	innovation, innovation variance, Kalman gain.
r_t, L_t	Durbin–Koopman backward recursion vector and transition.
$\alpha_{t\|H}$	smoothed (full-interval) state.
$\mathcal{W}$	COVID-as-missing window (the fixed three quarters 2020Q2–Q4).
GVAR coupling (Sec. 6)	
c, j	bloc (economy) indices; c receiver, j partner.
w_{cj}	trade weight of partner j in bloc c's stars, $\sum_j w_{cj} = 1$.
`ROWDEM`, `ROWINF`, `ROWFIN`	trade-weighted foreign demand, inflation, and financial "star" variables.
$s_{c,v,t}$	scenario pin-delta of bloc c's star v at horizon t.
$g\Delta_{p,t}$	deviation of US-determined global column p (oil, US10Y, USBAA, USeq).
x	stacked deviation state of the coupling fixed point.
M	solve-time coupling operator; $x = Mx + d$.
d	exogenous forcing (own-scenario pins) of the fixed point.
$\rho(M)$	spectral radius of M (multiplier strength; $< 1 \Rightarrow$ contractive).
Fiscal recursion (Sec. 7)	
d_t	gross public debt as a percent of GDP.
$rev_t, pexp_t$	revenue and primary expenditure (% of GDP); `FREV_GDP`, `FPEXP_GDP`.
i_t^{eff}	effective interest rate (`EFFI`, % p.a.).
sfa_t	stock-flow adjustment (`SFA`, % of GDP).
g_t^n	nominal GDP growth.
pb_t	primary balance $= rev_t - pexp_t$.
Long-run tilt & evaluation (Sec. 8, 13)	
$\bar{y}^{\text{LR}}$	user-specified long-run anchor (H-step moment of the tilted density).
θ	entropic-tilt multiplier (dual solution).
ESS	effective sample size of the tilted importance weights.
h	forecast horizon (quarters); $h \in \{1, 2, 4, 8\}$ in the horse-race.
t_0	forecast origin in the recursive out-of-sample scheme.
d_t (loss)	loss differential between two models at t (overloads d_t; context disambiguates).
CRPS	continuous ranked probability score (primary density metric).

2 BVAR Estimation

2.1 Reduced-Form VAR

Consider an n-variable VAR(p) in reduced form:

$$y_t = c + \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ is the vector of observables, $c \in \mathbb{R}^n$ is a vector of intercepts, $B_1, \ldots, B_p \in \mathbb{R}^{n\times n}$ are the autoregressive coefficient matrices, and $\Sigma \in \mathbb{R}^{n\times n}$ is the positive-definite innovation covariance matrix. In the US specification, $n = 41$ and $p = 3$, yielding a coefficient matrix $\hat{\beta} \in \mathbb{R}^{124\times 41}$ (with row 0 containing the intercept). The 41 US variables include the four US-determined *global commons* (`GLB_OIL`, `GLB_USLR`, `GLB_USBAA`, `GLB_USEQ`) that the GVAR layer of Section 6 carries into every other bloc, the four-variable fiscal block (Section 7), and the three trade-weighted `ROW*_US` star variables through which the US, now an endogenous GVAR bloc, receives Rest-of-World feedback.

2.2 Data Transformations

Variables enter the VAR in one of two forms:

- **Log-transformed** (`log`): the variable is modeled in log levels. Growth rates displayed to the user are quarter-over-quarter annualized:

$$g_t^{\text{q/q ann.}} = 400 \times \big(\ln Y_t - \ln Y_{t-1}\big). \tag{2}$$

  Year-over-year growth rates are computed as:

$$g_t^{\text{y/y}} = 100 \times \big(\ln Y_t - \ln Y_{t-4}\big). \tag{3}$$

- **Linear** (`lin`): the variable enters in levels (e.g., interest rates in percent, unemployment rate). No growth-rate conversion is applied.

2.3 Reporting Units

The display unit is chosen per variable for interpretability, independently of the estimation transformation:

- **National-accounts / activity flows** (GDP, consumption, investment, trade volumes) are shown as growth (q/q annualized or y/y).
- **Prices** (CPI/HICP/deflators) are shown as inflation.
- **Rates and ratios** (policy and market rates, unemployment, fiscal ratios) are shown as levels in percent / percentage points. Unemployment-rate uncertainty bands are floored at 0 (a rate

cannot be negative); policy and market rates are *not* floored, because negative nominal rates do occur.

- **Index / FX / equity / commodity series** (the trade-weighted dollar `FXTWM`, US equity `GLB_USEQ`, house prices `HP`, oil `PZTEXP/GLB_OIL`, the non-fuel commodity index `GSNE`, and the other blocs' `NEER_EA`, `NEER_UK`, `NEER_CA`, `REER_CN`, `EQ_CN`) are reported as the *index level* (e.g. $2010 = 100$) rather than as q/q growth: a trade-weighted exchange rate or equity index q/q-annualized is not an informative number, whereas its level is. These series remain `log` for estimation and are conditioned in the stored $100 \ln(\text{index})$ space; only the display is exponentiated back to the level, $\exp(\text{stored}/100)$, with a level-space uncertainty fan.

2.4 Hierarchical Minnesota Prior

The model is estimated following Giannone et al. (2015), who cast the Minnesota prior (Litterman, 1986; Doan et al., 1984) in a hierarchical framework with data-driven hyperparameter selection. The prior specification includes:

(a) The **Minnesota shrinkage** prior on B_ℓ, centered on the random-walk or white-noise prior for each variable (depending on the variable's stationarity properties). The tightness parameter λ controls the overall degree of shrinkage toward the prior mean.

(b) The **sum-of-coefficients** prior (Doan et al., 1984), which shrinks toward the belief that a unit increase in every variable's level produces a unit increase in its forecast, imposing a form of cointegration.

(c) The **single-unit-root** prior, which constrains the system to have at most one common stochastic trend.

The hyperparameters governing the tightness of each prior component are treated as additional parameters and selected by maximizing the marginal likelihood, as advocated by Giannone et al. (2015).

Posterior draws are obtained via Markov chain Monte Carlo (MCMC). The estimator runs a long chain with burn-in, retaining the posterior mean $\hat{\beta}$ and $\hat{\Sigma}$ for the deterministic conditional forecasting engine. For the browser the chain is *thinned* to $D = 500$ retained draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^{D}$, the count shipped in every bloc's `companion_draws.json`, used for uncertainty quantification via the simulation smoother (Section 4.6). The thinned export keeps the per-bloc draw artifact small enough to load on demand (the US file is $\approx 34\,\text{MB}$ at $D = 500$, $n = 41$, $p = 3$).

2.5 Per-variable own-lag prior mean: random walk vs. stationary

The Minnesota/GLP prior is centered, for each equation, on a univariate model in which the variable's own first lag carries a prior mean of δ_i and every other coefficient a prior mean of zero (Litterman, 1986; Doan et al., 1984; Giannone et al., 2015). The *default* choice $\delta_i = 1$ encodes a random-walk (`RW`) prior belief—"the best forecast of the level is the current level"—which is appropriate for variables that are (near-)integrated in the estimation space (log real activity, log price levels, nominal yields). For a variable that is genuinely *stationary*—a mean-reverting ratio or rate that fluctuates around a constant level—the random-walk center is the wrong null: it shrinks the dynamics toward a unit root and lets the variable *drift* at long horizons rather than return to its historical mean. For such variables the tool sets the own-lag prior mean to $\delta_i = 0$, a stationary white-noise (`WN`) prior whose null model is "the best forecast is the unconditional mean (the equation constant), not the last observation."

The mechanism (`pos`/`stationary`). The choice is implemented exactly once, at the point where the prior mean of the coefficient matrix is built. In `BVAR/MFBVAR/GLPreplicationWeb/bvarGLPmf.m` the own-first-lag prior block is the diagonal matrix $\texttt{diag}(\texttt{diagb})$ with $\texttt{diagb} = \mathbf{1}_n$ by default; the line

$$\texttt{diagb}(\texttt{pos}) = 0$$

zeroes the own-lag prior mean for every variable index in the vector `pos`, so those equations are shrunk toward white noise instead of a random walk. The index set is threaded into the sampler as the third positional argument `stationary` of `BVAR/MFBVAR/cgl_bvarV5.m` (internally `pos = stationary`), which forwards it as the `'pos'` option to `bvarGLPmf`. The estimators build the set directly from the per-variable `prior` tag in each bloc's specification,

$$\texttt{pos} \;=\; \texttt{find}(\texttt{prior} == \text{‘}\texttt{WN}\text{’}),$$

so a variable opts in to the stationary prior simply by carrying `prior:"WN"` in its spec (see `pipeline/export_bvar_gvar24_loop.m` and `pipeline/export_bvar_euro_fsap.m`). For the 44 non-fiscal blocs the set is empty and the call is byte-identical to the historical all-`RW` behaviour.

Prior, not constraint. Crucially $\delta_i = 0$ sets only the prior *mean*; it does not *impose* stationarity. The posterior own-lag coefficient is the usual GLP balance of prior and likelihood, governed by the estimated tightness λ (itself selected by the marginal-likelihood maximization, Section 2.4). A genuinely persistent series whose data favour a near-unit root will still produce a large posterior own-lag coefficient despite the white-noise center; the `WN` prior simply removes the *built-in* drift that the random-walk center would otherwise inject at the long end. This is the mechanism that makes the fiscal GDP-ratios mean-revert (Section 7.1).

3 Companion-Form State Space

To apply the Kalman filter, the VAR(p) in (1) is cast in companion form as a first-order state-space model (see, e.g., Durbin and Koopman, 2012, ch. 3).

3.1 State Vector

Define the np-dimensional state vector:

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}. \tag{4}$$

3.2 Transition Equation

The state evolves according to:

$$\alpha_t = T\,\alpha_{t-1} + d + R\,\eta_t, \qquad \eta_t \sim \mathcal{N}(0, \Sigma), \tag{5}$$

where the $np \times np$ transition matrix T, the np-dimensional intercept vector d, and the $np \times n$ selection matrix R are:

$$T = \begin{pmatrix} B_1' & B_2' & \cdots & B_{p-1}' & B_p' \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \quad d = \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad R = \begin{pmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{6}$$

Remark 1. The B_ℓ' blocks in T are transposed relative to (1) because the coefficient matrix $\hat{\beta}$ is stored as $(1 + np) \times n$, with row 0 containing the intercept c and rows $1, \ldots, np$ containing the stacked lag coefficients. Specifically, $T(i, j) = \hat{\beta}(j + 1, i)$ for $0 \le i < n$ and $0 \le j < np$.

3.3 Observation Equation

The observation equation extracts the contemporaneous variables:

$$y_t = Z\,\alpha_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, H), \tag{7}$$

where $Z = \big(I_n \mid 0_{n \times n(p-1)}\big)$ and $H = \sigma_\varepsilon^2 I_n$ with $\sigma_\varepsilon^2 = 10^{-12}$ (a negligible measurement error included for numerical stability of the Kalman filter recursions).

3.4 State-Noise Covariance

The $np \times np$ state-noise covariance matrix is:

$$Q = R\,\Sigma\,R' = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}. \tag{8}$$

4 Kalman Filter and Smoother

4.1 Conditioning via Partial Observations

The key insight underlying conditional forecasting in a state-space framework, following Bańbura et al. (2015), is that conditioning on variable i at horizon h is equivalent to observing $y_{T+h,i}$ in the Kalman filter. For each forecast period $t = T+1, \ldots, T+H$, the observation vector is partitioned into:

- **Conditioned variables:** $y_t^{(o)}$, with known target values specified by the user.
- **Free variables:** $y_t^{(m)}$, treated as missing observations (coded as `NaN`).

The Kalman filter processes only the observed subset at each time step, adapting Z_t, H_t, and the innovation v_t accordingly.

4.2 Forward Pass: Kalman Filter

Algorithm: Kalman Filter with Partial Observations. **Input:** T, Z, Q, H, d; initial state α_0, covariance P_0; partial observations $\{y_t^{(o)}\}_{t=1}^{H}$.
For $t = 1, \ldots, H$:

$$\alpha_{t|t-1} = T\,\alpha_{t-1|t-1} + d \qquad \text{(state prediction)} \tag{9}$$
$$P_{t|t-1} = T\,P_{t-1|t-1}\,T' + Q \qquad \text{(covariance prediction)} \tag{10}$$

If observations exist at time t, let $Z_t^{(o)}$ be the rows of Z for observed variables, then:

$$v_t = y_t^{(o)} - Z_t^{(o)}\,\alpha_{t|t-1} \qquad \text{(innovation)} \tag{11}$$
$$F_t = Z_t^{(o)}\,P_{t|t-1}\,(Z_t^{(o)})' + H_t^{(o)} \qquad \text{(innovation variance)} \tag{12}$$
$$K_t = P_{t|t-1}\,(Z_t^{(o)})'\,F_t^{-1} \qquad \text{(Kalman gain)} \tag{13}$$
$$\alpha_{t|t} = \alpha_{t|t-1} + K_t\,v_t \qquad \text{(state update)} \tag{14}$$
$$P_{t|t} = P_{t|t-1} - K_t\,Z_t^{(o)}\,P_{t|t-1} \qquad \text{(covariance update)} \tag{15}$$

If no observations exist at time t: $\alpha_{t|t} = \alpha_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$.

The matrix inversion F_t^{-1} is computed via LU decomposition with partial pivoting; singularity is flagged when the pivot falls below 10^{-15}.

4.3 Backward Pass: Durbin–Koopman Smoother

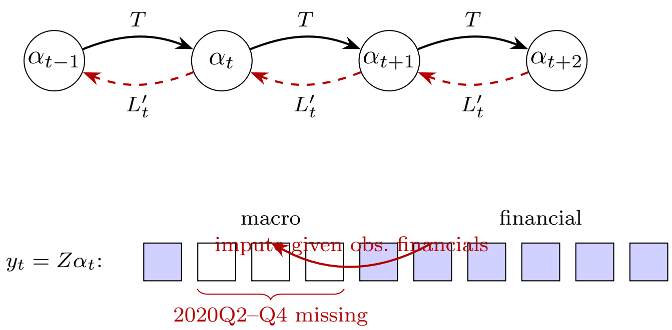

Figure 2: Companion-form state space with the Durbin–Koopman filter/smoother under the COVID-as-missing treatment. States α_t (Eq. (4)) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (9)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L_t' r_t$ (dashed, Eq. (17)). In the observation row $y_t = Z\alpha_t$ (Eq. (7)), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

The smoother refines the filtered estimates using all future information, yielding $\alpha_{t|H}$ for all t (Durbin and Koopman, 2002, 2012).

Algorithm: Durbin–Koopman Fixed-Interval Smoother. **Input:** Filtered quantities $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, Z_t^{(o)}$
Initialize $r_H = 0$. For $t = H, H-1, \ldots, 1$:

If observations exist at time t:

$$L_t = T' \left(I - P_{t|t-1} (Z_t^{(o)})' F_t^{-1} Z_t^{(o)}\right), \tag{16}$$

$$r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L_t' r_t. \tag{17}$$

Otherwise: $r_{t-1} = T' r_t$.
The smoothed state is:

$$\alpha_{t|H} = \alpha_{t|t-1} + P_{t|t-1} r_{t-1}. \tag{18}$$

Simulation smoother (draws, not just means). For uncertainty quantification and for the COVID-as-missing data augmentation (Section 9.4) the tool needs *draws* of the missing/conditioned states, not only their smoothed mean. The Durbin and Koopman (2002) simulation smoother obtains a draw $\tilde{\alpha}$ from the conditional distribution $p(\alpha \mid y^{(o)})$ by the elegant "mean-correction" device: simulate an unconditional path, smooth it, and add the smoothed *discrepancy* back to the unconditional draw. Algorithm 1 states the form used inside each `cgl_bvarV5` sweep, with partial (ragged + COVID-masked) observations handled by processing only the observed rows $Z_t^{(o)}$ at each date.

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep).

Input: system matrices T, Z, Q, H; partial observations $\{y_t^{(o)}\}_{t=1}^{H}$ (COVID-macro and ragged rows absent); current draw of (β, Σ).
Output: one draw $\tilde{\alpha}_{1:H}$ from $p(\alpha_{1:H} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for the missing y cells.

1: ▷ (1) Forward simulation of an unconditional path.
2: Draw α_0^+ from the initial state law; **for** $t = 1, \ldots, H$: draw η_t, ε_t and set $\alpha_t^+ = T\alpha_{t-1}^+ + \eta_t$, $y_t^+ = Z\alpha_t^+ + \varepsilon_t$.
3: Form the synthetic partial observations $y_t^{+(o)}$ on the *same* observed pattern as the data (drop the cells that are `NaN` in $y_t^{(o)}$).
4: ▷ (2) Kalman filter (Algorithm of Sec. 4.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$.
5: **for** $t = 1, \ldots, H$: run the prediction/update recursions on $Z_t^{(o)}$, storing $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, F_t\}$. ▷ at fully-missing t, no update
6: ▷ (3) Backward Durbin–Koopman smoothing recursion.
7: Set $r_H = 0$; **for** $t = H, \ldots, 1$:
8: **if** observed at t: $L_t = T'\big(I - P_{t|t-1}(Z_t^{(o)})'F_t^{-1}Z_t^{(o)}\big)$, $r_{t-1} = (Z_t^{(o)})'F_t^{-1}v_t + L_t'r_t$;
9: **else**: $r_{t-1} = T'r_t$.
10: smoothed discrepancy $\hat{\alpha}_t = \alpha_{t|t-1} + P_{t|t-1}r_{t-1}$.
11: ▷ (4) Mean-correction: add the discrepancy back to the unconditional draw.
12: $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed $\tilde{y}_t = Z\tilde{\alpha}_t$ at the missing cells.

Averaging $\tilde{\alpha}$ across Gibbs sweeps recovers the posterior-mean completed panel $\widehat{X}_c$ that Algorithm 3 feeds back into the stars; retaining the per-draw paths yields the predictive draw cloud `Dforecast` used for the uncertainty bands (Section 9.6) and for density scoring (Section 13.2).

4.4 Conditional Forecast Computation

The conditional forecast is computed as follows:

1. Initialize the state from the last p historical observations: $\alpha_0 = (y_T', y_{T-1}', \ldots, y_{T-p+1}')'$.
2. Construct the conditioning matrix $Y^* \in \mathbb{R}^{H\times n}$, where $Y_{t,j}^*$ is the target value for variable j at horizon t if conditioned, and `NaN` otherwise.
3. Run the Kalman filter (Algorithm 4.2) and smoother (Algorithm 4.3) with partial observations given by the non-`NaN` entries of Y^*.
4. Extract the observation block (first n elements) from each smoothed state $\alpha_{t|H}$.

4.5 Delta-Based Display

To ensure that zero user deviations produce exactly the stored baseline forecast, the tool computes both an unconditional smoother pass (with all entries of Y^* set to `NaN`) and a conditional pass. The displayed forecast is:

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \underbrace{\left(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}}\right)}_{\text{smoother delta}}, \tag{19}$$

where $\bar{y}^{\text{baseline}}$ is the pre-computed baseline forecast (posterior mean). This construction guarantees that the displayed forecast matches the stored baseline exactly when no conditioning constraints are imposed, avoiding small numerical discrepancies between the smoother's unconditional output and the stored baseline. The unconditional smoother result is cached and reused across conditioning changes.

For log-transformed variables, the smoother operates in log-level space, and the delta is scaled to annualized growth-rate space:

$$\Delta g_h = 4 \times \Delta \ell_h, \qquad \Delta \ell_h = \ell_h^{\text{cond}} - \ell_h^{\text{uncond}}, \tag{20}$$

where ℓ_h denotes the log-level forecast at horizon h.

4.6 Uncertainty Quantification

Forecast uncertainty bands are computed via the Durbin–Koopman *simulation smoother* (Durbin and Koopman, 2002), which generates draws from the posterior predictive distribution conditional on the user's constraints.

For each posterior draw $d = 1, \ldots, D$:

1. **Forward simulation:** Draw shocks $\eta_t^+ \sim \mathcal{N}(0, I_n)$ and compute a simulated path using the d-th posterior draw of the VAR coefficients:

$$\alpha_t^+ = T^{(d)}\, \alpha_{t-1}^+ + d^{(d)} + R\, L^{(d)}\, \eta_t^+, \tag{21}$$

   where $L^{(d)}$ is the lower Cholesky factor of $\Sigma^{(d)}$. The simulated observations are $y_t^+ = Z\, \alpha_t^+$.

2. **Deviation computation:** At conditioned entries, compute $y_t^* = Y_t^{\text{target}} - y_t^+$; at free entries, set $y_t^* =$ `NaN`.

3. **Smoother on deviations:** Run the Kalman smoother on $\{y_t^*\}$ with initial state $\alpha_0^* = 0$ and small initial covariance $P_0^* = 10^{-7} I_{np}$, producing adjustment paths $\hat{\alpha}_t^*$.

4. **Combine:** The conditional draw is $\tilde{\alpha}_t = \hat{\alpha}_t^* + \alpha_t^+$.

The 16th and 84th percentiles across D draws yield the reported 68% credible bands. The tool uses $D = 50$ in "Quick" mode and $D = 200$ in "Full" mode. The simulation smoother correctly integrates over both parameter uncertainty (through the posterior draws) and forecast uncertainty (through the simulated shocks).

5 Policy Counterfactuals

The counterfactual layer evaluates how macroeconomic outcomes change under an alternative policy stance. It operates *on top of* the conditional scenario described in Section 4: starting from the DK-smoothed scenario paths, the user specifies deviations in the policy rate or government spending, and the tool traces the implied effects through a DSGE model's impulse response functions before re-conditioning the full system through the BVAR.

5.1 DSGE Impulse Response Functions

Impulse response functions (IRFs) are drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). The MMB provides a library of pre-solved DSGE models with standardized variable names and multiple policy rule specifications. The tool includes IRFs from 169 models under 9 monetary policy rules:

Rule	**Reference**
Taylor	Taylor (1993)
SW	Smets and Wouters (2007)
CEE	Christiano et al. (2005)
GR	Gerdesmeier & Roffia (2004)
LWW	Levin, Wieland & Williams (2003)
OW08	Orphanides & Wieland (2008)
OW13	Orphanides & Wieland (2013)
Coenen	Coenen et al. (2012)
CMR	Christiano, Motto & Rostagno (2014)

Each IRF file provides the dynamic response of key macroeconomic variables—inflation, output gap, interest rate, consumption, investment, and others—to a one-standard-deviation monetary policy shock under the specified rule. Fiscal policy IRFs are available for 60 models that include a fiscal sector.

5.2 Monetary Policy Shock Inversion

Given a user-specified interest-rate adjustment path $\Delta r = (\Delta r_1, \ldots, \Delta r_H)'$, the tool solves for the structural monetary policy shock sequence $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_H)'$ via the lower-triangular Toeplitz system:

$$\underbrace{\begin{pmatrix} h_r(0) & 0 & \cdots & 0 \\ h_r(1) & h_r(0) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ h_r(H-1) & h_r(H-2) & \cdots & h_r(0) \end{pmatrix}}_{\mathcal{H}_r \in \mathbb{R}^{H\times H}} \varepsilon = \Delta r, \tag{22}$$

where $h_r(s)$ is the impulse response of the interest rate at horizon s. Since $\mathcal{H}_r$ is lower-triangular, the system is solved by forward substitution:

$$\varepsilon_t = \frac{\Delta r_t - \sum_{j=0}^{t-1} h_r(t-j)\,\varepsilon_j}{h_r(0)}, \qquad t = 0, 1, \ldots, H-1. \tag{23}$$

The implied responses of inflation and output are:

$$\Delta\pi_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_\pi(t-j)\,\varepsilon_j, \qquad \Delta y_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_y(t-j)\,\varepsilon_j. \tag{24}$$

An analogous procedure applies to fiscal policy shocks, using the government-spending IRFs (h_g, h_π^g, h_y^g, h_r^g) from the DSGE models in the MMB that include a fiscal sector.

5.3 Three Conditioning Modes

After computing the DSGE-implied deltas, the counterfactual is re-conditioned through the BVAR to enforce cross-variable consistency. Three modes govern what is held fixed versus free in the second-stage DK smoother pass:

(a) Hold-targets: All user-conditioned variables are pinned at their scenario values; the interest rate is pinned at the scenario rate plus the adjustment. Remaining variables are free and determined by the DK smoother:

$$Y_t^{\mathrm{cf}}(j) = \hat{y}_{T+t,j}^{\mathrm{cond}}, \qquad \text{for } j \in \mathcal{C}, \tag{25}$$

$$Y_t^{\mathrm{cf}}(r) = \hat{y}_{T+t,r}^{\mathrm{cond}} + \Delta r_t, \tag{26}$$

$$Y_t^{\mathrm{cf}}(j) = \texttt{NaN}, \qquad \text{otherwise.} \tag{27}$$

(b) Let-adjust: Only the policy instrument(s) are conditioned; all other variables, including inflation and output, are free and adjust endogenously through the BVAR covariance structure.

(c) IRF-consistent (default): The interest rate is pinned at the scenario rate plus the adjustment, and inflation and output are pinned at the scenario values plus the DSGE-implied deltas from (24):

$$Y_t^{\mathrm{cf}}(\pi) = \hat{y}^{\mathrm{cond}}_{T+t,\pi} + \Delta\pi_t^{\mathrm{IRF}}, \tag{28}$$
$$Y_t^{\mathrm{cf}}(y) = \hat{y}^{\mathrm{cond}}_{T+t,y} + \Delta y_t^{\mathrm{IRF}}, \tag{29}$$
$$Y_t^{\mathrm{cf}}(r) = \hat{y}^{\mathrm{cond}}_{T+t,r} + \Delta r_t. \tag{30}$$

The remaining variables are determined by the DK smoother, producing responses consistent with both the DSGE-identified monetary transmission and the BVAR covariance structure.

In all modes, the counterfactual display uses the same delta construction as (19):

$$\hat{y}^{\mathrm{cf,display}}_{T+h} = \hat{y}^{\mathrm{scen,display}}_{T+h} + f\big(\hat{y}^{\mathrm{cf,smooth}}_{T+h} - \hat{y}^{\mathrm{scen,smooth}}_{T+h}\big), \tag{31}$$

where $f(\cdot)$ applies the $\times 4$ scaling from (20) for log-transformed variables.

5.4 Optimal Policy: Barnichon–Mesters

Following Barnichon and Mesters (2023), the tool solves for the optimal monetary policy shock sequence that minimizes a quadratic loss function over deviations from target:

$$\mathcal{L} = \sum_{t=1}^{H} \Big[\lambda_\pi \big(\pi_t - \pi^*\big)^2 + \lambda_y \big(y_t - y^*\big)^2\Big], \tag{32}$$

subject to the linear mapping from shocks to outcomes given by the DSGE impulse responses.

Define the Toeplitz matrices $\mathcal{H}_\pi, \mathcal{H}_y \in \mathbb{R}^{H\times H}$ analogously to (22) using the inflation and output-gap IRFs. Stacking the weighted system:

$$\underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\,\mathcal{H}_\pi \\ \sqrt{\lambda_y}\,\mathcal{H}_y \end{pmatrix}}_{\tilde{\mathcal{H}}\in\mathbb{R}^{2H\times H}} \varepsilon = \underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\,\Delta\pi^{\mathrm{target}} \\ \sqrt{\lambda_y}\,\Delta y^{\mathrm{target}} \end{pmatrix}}_{\tilde{b}\in\mathbb{R}^{2H}}, \tag{33}$$

the optimal shock sequence is obtained via Tikhonov-regularized least squares:

$$\varepsilon^* = \big(\tilde{\mathcal{H}}'\tilde{\mathcal{H}} + \alpha I_H\big)^{-1}\tilde{\mathcal{H}}'\,\tilde{b}, \tag{34}$$

with regularization parameter $\alpha = 10^{-6}$. The implied policy rate path is $\Delta r_t^* = \sum_{j=0}^{t} h_r(t-j)\,\varepsilon_j^*$, which is then passed to the IRF-consistent conditioning mode (Section 5.3c) to obtain the full counterfactual.

The user controls the relative weight on price stability versus full employment via a slider that sets $\lambda_\pi/(\lambda_\pi + \lambda_y)$.

6 Global VAR (GVAR) and Global Commons

The single-country engine of Sections 2–5 generalizes to a multi-country setting in which a conditioning imposed on one economy spills over to all others. The tool implements a *Global VAR* (GVAR) layer in the tradition of Pesaran et al. (2004) and Dees et al. (2007), comprising the United States together with the euro area as a single aggregate bloc and 31 further Rest-of-World (RoW) blocs—33 coupled blocs in all—that span approximately 90% of global GDP. The decisive implementation choice is that *cross-country coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated single-country BVAR, and the spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Two further channels distinguish the current version: the United States is now a *fully endogenous* coupled bloc—it drives the rest of the world *and* receives feedback from it, rather than acting as a frozen anchor—and four US-determined *global commons* (oil, the long US rate, US corporate credit, and US equity) are carried by every bloc.

6.1 Architecture: an endogenous US and 32 partner blocs

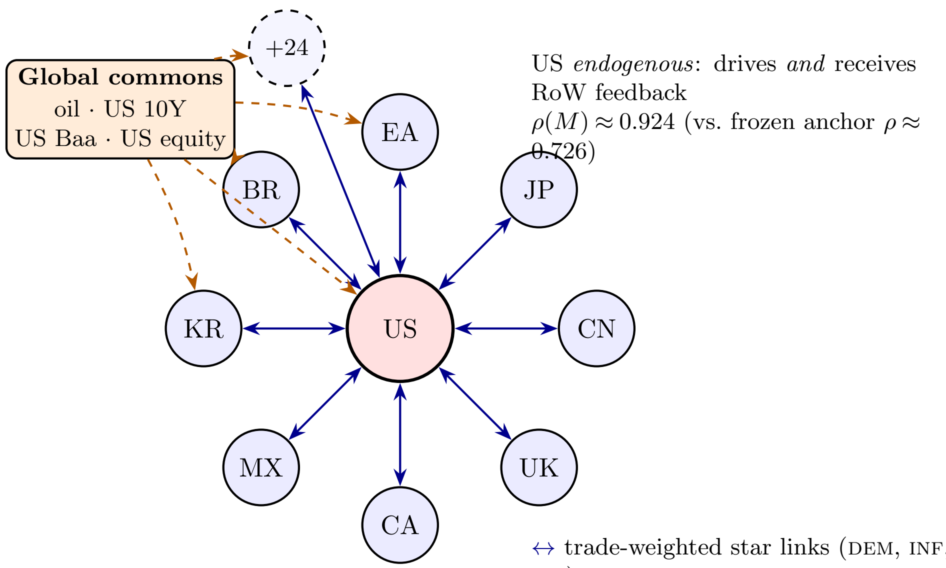

Figure 3: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 32 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 24 further blocs) through *bidirectional*, trade-weighted "RoW star" links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically (1 : 1) to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator's spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only) to $\rho(M) \approx 0.924$ at the full 33-bloc endogenous-US configuration (the value stored in the shipped `coupling_default.json.gz`; it moves slightly with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 6.1–6.4).

The bloc set is {US, EA, JP, UK, CA, CN, CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, T (33 blocs: the US, the euro-area aggregate `EA`, and 31 further blocs). The euro area enters as a *single aggregate* bloc rather than as its individual members: a fully disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$), and aggregating the euro area into one bloc restores stationarity. The 17 euro-area member states are therefore retained only as *standalone* single-country models outside the coupled system (Section 9.2), not as GVAR blocs.

The US is now endogenous, not a frozen anchor. In earlier versions the US was treated as a *closed anchor*, exogenous in the Dees–PSW sense (Dees et al., 2007): it carried no foreign variables, was solved once, and never iterated in the cross-country fixed point. The current US model instead carries its own three trade-weighted RoW *star* variables (`ROWDEM_US`, `ROWINF_US`, `ROWFIN_US`) built from a US trade-weights row, so the US is a *fully coupled* bloc with $n = 41$ (its domestic and fiscal variables, the global commons of Section 6.6 as its own endogenous variables, and the three US star variables). It both drives every partner—through its weight in their trade-weighted aggregates and through the global commons—*and* receives partner feedback through its own stars. The US is the first entry of the coupled set and iterates inside the cross-country fixed point alongside everyone else; there is no longer a privileged frozen bloc. This RoW→US feedback is the qualitative gain of the endogenous treatment, and it is also what makes the complete solve more expensive (Section 6.5).

The topology is **not hardcoded**: it self-configures from the deployed model specifications. The US is treated as endogenous if and only if its model carries the three `ROW*_US` star variables *and* it has a row in the trade-weights file; absent either, the engine falls back to the legacy frozen-anchor solve, which remains byte-identical. More generally, a bloc is an active GVAR member if and only if its `spec.json` carries variables tagged `block: 'row'` *and* it has a row in the trade-weights file (`weights_24bloc.json`; the filename is historical—the file now lists all 33 coupled blocs, the US row included). At load, the engine derives each bloc's GDP, price, and short-rate columns from its specification (mirroring the estimator's column-resolution order) and reads its RoW-star indices, so that bloc indices are never assumed by position. Adding an economy to the GVAR therefore requires only shipping its estimated model with the appropriate variable tags and a weights row.

6.2 Trade-weighted star variables

Every coupled bloc—now including the United States—consists of its domestic VAR augmented by three trade-weighted *star* variables that summarize the foreign environment, following the weak-exogeneity device of Pesaran et al. (2004):

- `ROWDEM`: foreign demand, a trade-weighted index of partner real-GDP growth;
- `ROWINF`: foreign inflation, the analogous index on partner price levels;
- `ROWFIN`: foreign financial conditions, a trade-weighted average of partner short-term policy rates.

Let $\mathcal{W}_c \subseteq \{\text{partners of } c\}$ collect the partners that contribute to a given star at quarter t, with w_{cj} the (renormalized) trade weight of partner j in receiver c. For the activity and price stars, the partner level is read in $100\ln$ space and the star is the annualized growth index

$$\mathrm{ROWDEM}_{c,t} = 4\sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\mathrm{gdp}}_{j,t} - \ell^{\mathrm{gdp}}_{j,t-1}\right), \qquad \mathrm{ROWINF}_{c,t} = 4\sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\mathrm{price}}_{j,t} - \ell^{\mathrm{price}}_{j,t-1}\right), \tag{35}$$

where the scaling $4\,\Delta(100\ln\cdot)$ is the quarter-over-quarter annualized convention of (2), and the financial star is the level aggregate

$$\mathrm{ROWFIN}_{c,t} = \sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\, r_{j,t}. \tag{36}$$

The weights are *per-component* renormalized over the partners that are present and eligible at each t, $\tilde{w}_{cj} = w_{cj}\left(\sum_k w_{ck}\right)/\left(\sum_{k\in\mathcal{W}_c} w_{ck}\right)$, so that the mass of any absent or excluded partner is reabsorbed and the active weights still sum to one. Three eligibility rules mirror the estimator exactly: rate-less blocs (Switzerland, Norway) carry no own short rate and are dropped from partners' `ROWFIN` (but retained in `ROWDEM`/`ROWINF`); the hyperinflation bloc (Argentina) is dropped from partners' `ROWINF` and `ROWFIN` (retained in `ROWDEM`); and China's GDP is reported in *nominal* terms, so its activity contribution is deflated by *subtraction* in log space, $\ell^{\mathrm{gdp}}_{\mathrm{CN}} - \ell^{\mathrm{price}}_{\mathrm{CN}} = 100\ln(\mathrm{NGDP}/P)$, rather than by division.

6.3 Trade weights

The bilateral weights are constructed from IMF International Merchandise Trade Statistics (IMTS) and shipped as `weights_24bloc.json` (a historical filename: the file now holds the full roster), which lists the 33 blocs together with 33 weight rows—one per bloc, the US included, now that the US is endogenous—each summing to one (zero diagonal). At load, the live weight matrix W is renormalized over the *active* bloc set, so a deployment that activates a subset of the candidate blocs remains internally consistent.

6.4 Coupling at solve time, not estimation

Because each bloc is estimated independently, there is no joint, block-exogenous GVAR system; the cross-country linkage is imposed when the scenario is solved. The key observation is that, with coefficients fixed at their posterior mean $(\hat{\beta}, \hat{\Sigma})$, the conditional-forecast operator of Section 4 is *affine* in the conditioning values (the Kalman filter and Durbin–Koopman smoother are linear in the pinned data). The *baseline* forecast of each bloc is therefore data-driven and computed exactly once, via an all-`NaN` (unconditional) smoother pass; only the scenario *deviation* couples across blocs.

Because the US is now endogenous it is no longer held outside the fixed point, and the compact deviation state is correspondingly *augmented*. It collects (i) the star pin-deltas $s_{c,v,t}$ of *every* coupled bloc c—the US included—for star variable $v \in \{\mathrm{DEM}, \mathrm{INF}, \mathrm{FIN}\}$ at horizon t, and (ii) the per-period

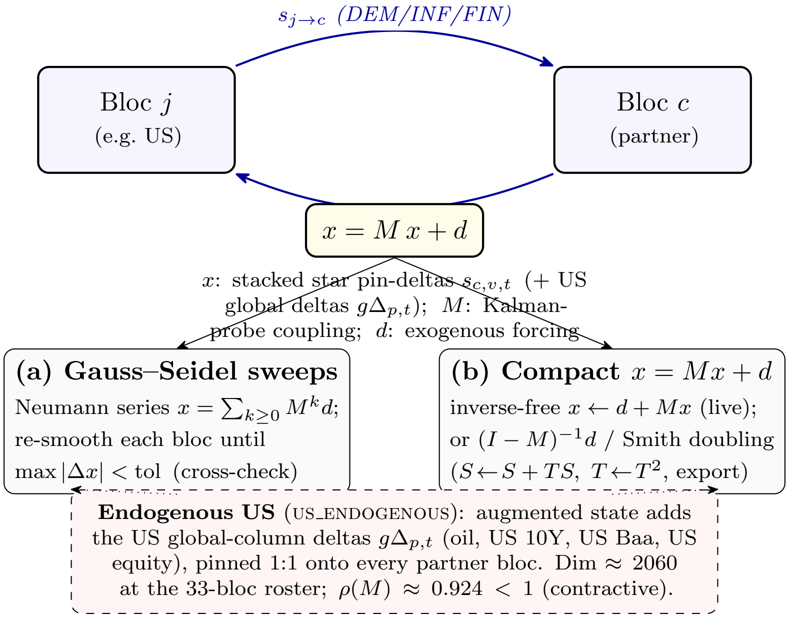

Figure 4: Solve-time cross-country coupling as the affine fixed point $x = M\,x+d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss–Seidel/Neumann sweeps through the Durbin–Koopman smoother, used as the cross-check, and (b) the compact M-state recursion, solved inverse-free by the Neumann fixed-point $x \leftarrow d + Mx$ on the production path (the explicit $(I - M)^{-1}d$ / Smith doubling are reserved for the small frozen-anchor case and the export-time precompute, since the in-browser $O(\dim^3)$ inverse is prohibitive at dim $\approx$ 2060). With the US endogenous the augmented state has dimension $\approx$ 2060 and spectral radius $\rho(M) \approx 0.924 < 1$, so the iteration contracts and the two routes agree to $\sim 10^{-9}$.

deviation of each US global column $g\Delta_{p,t}$, which under the frozen-anchor design was a constant and is now itself part of the unknown. Stacking these into a vector x, the deviation system is the linear fixed point

$$x = M\,x + d, \tag{37}$$

where d is the exogenous forcing (each bloc's own scenario pins moving its linked series at zero partner and global delta) and M now carries three coupling families: star $\leftarrow$ star (receiver c's star responds to a partner j's free-variable path—with the US participating both as partner and as receiver), the US-global definition rows (a global column's deviation as the US's own response to its star pins), and star $\leftarrow$ global (a partner's star responds to its own global pin). The US-determined globals therefore live *inside* M rather than in d, which is precisely the new RoW→US feedback loop. The augmented state has dimension $\approx$ 2060 at the full 33-bloc roster.

The fixed point (37) is solved two equivalent ways:

(a) **Gauss–Seidel / Neumann sweeps** (the cross-check) sum the Neumann series $x = \sum_{k\geq 0} M^k d$ by re-conditioning each bloc in turn through the Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) until the maximum level change falls below tolerance.

(b) **Direct closed form**

$$x = (I - M)^{-1} d, \tag{38}$$

after which one final conditional-forecast pass per bloc reconstructs the full set of free variables consistent with the converged pin-deltas.

At the full 33-bloc endogenous-US configuration the spectral radius is $\rho(M) \approx 0.924 < 1$, so the Neumann series converges and $(I - M)$ is nonsingular; the two solvers agree to $\sim 10^{-9}$. (The legacy frozen-anchor RoW-only operator had a smaller radius, $\rho(M) \approx 0.726$; closing the RoW$\to$US loop raises the coupling but leaves the system comfortably contractive.)

Algorithm 2 states both solve paths (`app/src/lib/compute/gvarSolve.ts`). The coupling operator M is never formed by writing down a joint VAR; each of its columns is *probed* by perturbing one pinned cell and reading the affine conditional-forecast response of every linked bloc through the Durbin–Koopman smoother—so M encodes the same per-bloc Kalman operator that produces the baseline, restricted to the deviation.

A practical consequence drives the deployment. The operator M depends on the estimated models and on the conditioning *structure* (which cells are pinned) but *not* on the pinned *values*, so it can be assembled once and reused for every value change during interactive dragging. Its dense inverse is, however, expensive: at the augmented endogenous dimension (≈ 2060) the naive in-browser inversion would take tens of minutes (the cubic inverse is of order three-quarters of an hour), making a cold build infeasible even in a Web Worker. The tool therefore **ships a precomputed coupling artifact** (`coupling_default.json.gz`) containing M, $(I - M)^{-1}$, and $\rho(M)$ for the default (empty) pin structure, with the inverse computed offline by a high-performance BLAS routine. The artifact is large (the dense 2060×2060 inverse), so it is shipped *gzipped* (about 30 MB compressed) and decompressed in the browser via the native `DecompressionStream` API at load. This artifact is mandatory, not an optimization: hydrating it reduces the first complete solve from the tens-of-minutes cold build to a few seconds. A model-signature gate guards against using a stale inverse after any re-estimation; a genuinely novel pin structure misses the cache and falls back to an inverse-free Neumann solve (the in-browser cubic inverse is never built live).

6.5 Two-tier solve: live preview vs. the full fixed point

The endogenous US makes the *complete* cross-country solve materially more expensive than the legacy frozen-anchor solve, because placing or moving a US pin re-probes the (large) US bloc, the heavy step. To keep the interface reactive the tool exposes **two solve tiers**:

- **Direct preview** (the default while dragging). A fast, first-order approximation that propagates the scenario through the pre-warmed local per-bloc elasticities—one cross-country hop, without the full multiplier feedback—so the propagated paths update live as the user drags a pin. The per-bloc response kernels are cached by model signature and own-pin structure, so a value-only re-drag (same pinned cells) reuses them and re-solves cheaply; a *structure* change

Algorithm 2 Solve-time coupling fixed point $x = Mx + d$ (Gauss–Seidel/Neumann and direct $(I - M)^{-1}$; `gvarSolve.ts`).

Input: per-bloc posterior-mean models $\{(\hat{\beta}_c, \hat{\Sigma}_c)\}$; scenario pins; trade weights W; `US_ENDOGENOUS` flag; tolerance TOL.

Output: converged deviation state x and per-bloc scenario deltas `delta`$[c]$; sweep count; $\rho(M)$.

$\triangleright$ Forcing. Build d: each bloc's own scenario pins, evaluated at *zero* partner-star and global deltas (one DK conditional-forecast pass per pinned bloc).

$\triangleright$ Coupling families in M (endogenous-US augmented state). star←star (a receiver's star responds to a partner's free path; the US is both partner and receiver); US-global rows ($g\Delta$ as the US's own response to its star pins); star←global (a partner star responds to its own global pin).

Path A — Gauss–Seidel / Neumann sweeps (inverse-free).

  $x \leftarrow d$

  repeat (sweep): **for each** coupled bloc c (US included **if** `US_ENDOGENOUS`): re-condition c through the DK smoother on the current partner-star and global deltas $\Rightarrow$ update x's block for c.

    $\triangleright$ this sums the Neumann series $x = \sum_{k \geq 0} M^k d$ term by term.

  until $\max |\Delta x| <$ TOL; record ρ_{emp} from the geometric residual decay.

Path B — direct closed form (cached).

  assemble M by probing columns; **if** a precomputed $(I - M)^{-1}$ matches the model signature & pin structure, hydrate it; **else** fall back to Path A (the dense in-browser inverse is never built live).

  $x \leftarrow (I - M)^{-1} d$; one final per-bloc DK pass reconstructs all free variables consistent with the converged pin-deltas.

$\triangleright$ Cross-check. The two paths agree to the verified tolerance ($\max |\Delta| < 10^{-4}$ across all blocs in the closed-form vs. iterative regression tests, with the iterative reference itself converged to 10^{-7}), confirming $\rho(M) < 1$; report $\rho(M)$ via power iteration on M (in practice on M^2, see Sec. 6.5).

return x, `delta`$[c]$, sweep count, $\rho(M)$.

(pinning a new cell) re-probes the US bloc and is the slower direct step (a few seconds), during which a small "computing…" pill is shown so the wait does not look like a freeze.

- **Full GVAR** (on demand). The complete fixed point of (37)—all feedback loops, including RoW→US—computed when the user clicks the **Solve full GVAR** button. This consumes the hydrated $(I - M)^{-1}$ artifact and takes a few seconds.

A badge above the charts reports which tier produced the current result—an amber "GVAR DIRECT preview" versus a blue "GVAR full solve"—together with the sweep count, the spectral radius ρ, and the solve time. When a DIRECT preview is on screen, the **Solve full GVAR** button is offered to upgrade it to the complete fixed point. Drag-release dispatch and debouncing keep the preview channel from firing mid-drag, and the heavy coupling artifact is stripped from the worker messages on the direct path (it is only needed for the full solve). For the legacy frozen-anchor topology—or any deployment where the US is not endogenous—both tiers resolve to the same already-fast cached closed form, so the two-tier distinction is invisible there.

6.6 Global commons: US-determined common shocks

The headline addition of version 2.0 is a set of four *global common* variables that are determined by the United States but carried by *every* bloc:

- `GLB_OIL` — the spot oil price (WTI);
- `GLB_USLR` — the US 10-year Treasury yield;
- `GLB_USBAA` — the US Baa corporate-bond yield (a credit-spread proxy);
- `GLB_USEQ` — US equity prices.

In the US bloc these enter as ordinary *endogenous* variables; in each partner bloc the same four identifiers (matched one-to-one by *name*, never by position) are carried as weakly-exogenous commons that the solver *pins*.

The pinning is by *deviation*, not by level. Each partner bloc c sets each global column to its own baseline plus the US deviation of the matching global,

$$\widehat{\mathrm{GLB}}_{c,t} = \bar{\mathrm{GLB}}_{c,t} + \big(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \bar{\mathrm{GLB}}_{\mathrm{US},t}\big), \tag{39}$$

where $\bar{(\cdot)}$ denotes the bloc's own unconditional baseline. This is a load-bearing design choice. Each RoW bloc estimated its *own* baseline path for the globals as weakly-exogenous regressors, and those baseline levels differ from the US's by tens of 100 ln units (for example, of order +17, +70, and +28 for the euro area, Japan, and China on oil). Pinning the US *level* would inject that entire baseline gap as a spurious deviation the instant a bloc is selected, even under a zero scenario. Pinning the US *deviation* on top of each bloc's own baseline instead yields exactly zero spillover at a zero scenario and an *identical* common shock under conditioning—precisely mirroring the star convention $\bar{\mathrm{RoW}}$ + partner-delta of (35)–(36). A US oil pin therefore propagates the same oil deviation to all 32 partner blocs (verified to within 10^{-9}), while each bloc's domestic variables respond through its *own* estimated coefficients.

Mechanically, with the US now endogenous the US global *deviation* is itself an unknown of the fixed point rather than a frozen constant. The global columns therefore enter the *augmented* coupling operator M of Section 6.4: a US-global definition row writes each global's deviation as the US's own response to its star pins, and a star $\leftarrow$ global block routes each partner's global pin back into its stars. This is exactly the channel by which a partner shock now feeds back to the US. Indirect effects—a common shock moving a bloc's macroeconomy, which moves its stars, which spill to other blocs—flow through the same operator and are not double-counted. A bloc that carries no globals reduces the pin to a no-op, leaving the legacy globals-free solve byte-identical.

On the user interface the globals are *conditionable* on the US screen (where they are endogenous) and *read-only*, carried over from the US path, on every other bloc.

6.7 Relation to the fiscal and counterfactual layers

The GVAR layer composes with the single-country machinery rather than replacing it. The displayed scenario line for an active GVAR bloc is the cross-country propagated path of this section; for a standalone (non-GVAR) economy it is the single-country conditional forecast of Section 4. The US derived fiscal block (the debt-to-GDP accounting identity, whose empirical motivation is the fiscal reaction function of Bohn, 1998) and the DSGE policy counterfactual—including the Barnichon and Mesters (2023) optimal-policy problem—continue to operate on the US bloc's propagated path. Conditional forecasts throughout use the large-cross-section state-space method of Bańbura et al. (2015) with the hierarchical prior of Giannone et al. (2015), applied independently to each bloc and stitched together only through the linear deviation system (37).

7 Fiscal Blocks: Estimated Drivers and Debt Accounting

Six of the economies in the tool—the United States, the Euro Area, Japan, the United Kingdom, Canada, and China—carry a dedicated *fiscal block*. The block follows the stock-flow-consistent division of labour shared by the U.S. Congressional Budget Office's conditional-BVAR budget mechanics and the IMF Financial Programming (FPP) / debt-sustainability frameworks: the BVAR is used as a *macro-fiscal scenario generator* that jointly forecasts the macroeconomy together with a small number of fiscal *flow drivers*, while the public-debt stock and the headline reporting variables (primary and overall balance, interest, debt-to-GDP, gross financing needs) are *derived outside the VAR* by the government budget-constraint identity. No reduced-form equation is ever allowed to forecast the debt stock freely, because an unconstrained debt VAR is explosive: the debt-dynamics identity has a near-unit autoregressive root $(1 + i)/(1 + g^n)$, and small parameter errors compound into divergent debt paths. Imposing the identity by construction is exactly the discipline that the CBO and FPP/DSA frameworks demand, and it is the structure implemented in `app/src/lib/compute/fiscalAccounting.ts`.

7.1 Estimated Fiscal Drivers

The BVAR jointly estimates the macro block of Section 9 together with *four* fiscal flow drivers, all entering as linear (`lin`) variables:

- **Revenue** (`FREV_GDP`): general-government receipts as a percentage of GDP, an annual ratio.
- **Primary expenditure** (`FPEXP_GDP`): non-interest outlays as a percentage of GDP, an annual ratio.
- **Effective interest rate** (`EFFI`, % p.a.): the average rate the government actually pays on its outstanding stock, constructed as net interest divided by the lagged gross debt and annualized. Because the stock turns over only as maturing securities are refinanced, this effective rate lags market yields and is therefore distinct from—and estimated separately from—the policy rate and the market yields already in the macro block.

- **Stock-flow adjustment** (`SFA`, % of GDP): the deficit–debt discrepancy described below.

The effective rate `EFFI` is the estimated interest driver of the deployed block. (The parametrisation has changed over time: an early version carried Δ(debt/GDP) directly, and a later one carried debt service to GDP `DSVC`; the current block reverts to the effective rate.) The interest bill is then formed inside the accounting recursion as $int_t = i_t^{\text{eff}}\, d_{t-1}/100$.

The stock-flow adjustment as an estimated variable. The novelty of the current block is that the stock-flow adjustment (`SFA`) is now an *estimated* BVAR variable rather than a constant plug. It is the exact residual of the tool's own debt recursion—the part of the observed change in gross debt that the flow side (deficit plus the interest–growth "snowball") does *not* explain. Economically this seam arises because the headline deficit is measured on a federal-flows perimeter while the debt stock is gross debt (so intragovernmental holdings, valuation effects, and other below-the-line "other means of financing" operations move the two apart). It is constructed so that it closes the recursion to machine precision on history, and the pandemic quarters are set to missing (`NaN`) so the COVID-as-missing Gibbs sampler imputes them (Section 9.4) rather than letting the 2020 swing contaminate the estimate. Carrying `SFA` as a forecast variable means the debt-to-GDP *projection* now propagates its own `SFA` path forward (with dynamics and uncertainty bands) instead of defaulting the discrepancy to zero, which materially affects the high-debt blocs whose historical SFA is large and persistent. The historical SFA means (in % of GDP), which each bloc also ships as its `sfaDefault` fallback, are: US 1.75, EA 0.33, JP 2.23, UK 3.15, CA 3.19, CN 4.02.

Because all four drivers are ordinary BVAR variables, they inherit the full estimation machinery of Sections 2–4: the hierarchical Minnesota prior (Giannone et al., 2015), the COVID-as-missing data-augmentation Gibbs treatment, the companion-form Durbin–Koopman conditional smoother (Durbin and Koopman, 2002, 2012), and the Bańbura et al. (2015) conditioning-via-partial-observations construction. In the five Global-VAR fiscal blocs (EA, JP, UK, CA, CN)—and now in the endogenous US bloc itself—the fiscal drivers are part of the domestic block and propagate to the other economies through the shared Rest-of-World stars, exactly as the macro variables do (Pesaran et al., 2004; Dees et al., 2007).

A stationary prior for the fiscal ratios. The three GDP-ratio drivers—revenue `FREV_GDP`, primary expenditure `FPEXP_GDP`, and the stock-flow adjustment `SFA`—are mean-reverting shares that fluctuate around a roughly constant in-sample level; they are *not* integrated. Carrying them under the default random-walk own-lag prior ($\delta = 1$) is therefore the wrong null: it injects a built-in drift that lets the projected ratio wander away from its historical level at the long end of the horizon. These three drivers are accordingly tagged `prior:"WN"` in every fiscal bloc's specification and estimated with the *stationary* white-noise own-lag prior ($\delta = 0$) of Section 2.5: the estimator collects them via `pos = find(prior == 'WN')` and passes the set as the `stationary` argument of `cgl_bvarV5`, so `diagb(pos) = 0` centers their own-lag dynamics on mean reversion to the equation constant. The effective interest rate `EFFI`, by contrast, *retains* the random-walk prior: it is a slow-moving stock-average rate (net interest over lagged debt) whose level is genuinely persistent—it turns over only as maturing securities refinance—so a near-unit own-lag is the right prior null for it.

The empirical effect is exactly the intended one. Under the random-walk prior the U.S. revenue ratio was pinned near its end-of-sample value ($\sim 19\%$) and drifted from there; under the white-noise prior it now *reverts* to its $\sim 17.4\%$ in-sample mean over the projection, and the expenditure ratio and `SFA` likewise return to their historical means rather than carrying the last observation forward indefinitely. We emphasise (per Section 2.5) that this is a shift of the prior *center*, not a hard stationarity constraint: the posterior own-lag coefficient is still the GLP prior–likelihood balance at the estimated tightness λ, so a fiscal ratio whose data genuinely trend will still be allowed to do so—the `WN` prior only removes the mechanical drift that the random-walk center would otherwise build in. Because `pos` is empty for the 44 non-fiscal blocs, this change is a no-op (byte-identical output) everywhere except the six fiscal blocs.

Each fiscal bloc ships a `fiscal_meta.json` file that pins the column layout used by the accounting recursion. It carries the integer indices `idx` = {`gdp`, `deflator`, `revenue`, `primaryExp`, `effRate`, `sfa`} into the level-space forecast matrix—the first two locate the real-GDP and deflator log-levels that supply nominal growth, the next two locate the revenue and primary-expenditure drivers, `effRate` locates the estimated effective-rate column, and `sfa` locates the estimated stock-flow-adjustment column—together with the seed debt ratio d_0 and a fallback `sfaDefault` (e.g. the U.S. block has $d_0 = 122.57$, `idx.effRate` = 35, `idx.sfa` = 36, and `sfaDefault` = 1.75; the U.K. block has $d_0 = 96.24$, `idx.effRate` = 18, `idx.sfa` = 19). The presence of `idx.sfa` is the switch that selects the per-quarter estimated SFA: when it is set, `computeFiscalBlock` reads the forecast SFA off that column so the debt projection carries the discrepancy forward with its own dynamics and bands; when it is unset, the recursion falls back to the constant `sfaDefault` (or zero). A legacy block that instead carried an `idx.debtService` column would read that column directly as the period interest flow and back out the effective rate; the current effective-rate blocks do not.

7.2 The Budget-Identity Recursion

Let d_t denote the debt-to-GDP ratio (in percent), i_t^{eff} the effective rate (percent per annum), and g_t^n nominal GDP growth (percent per quarter). All flows below are annual ratios in percent of GDP. Nominal growth is read directly off the level-space forecast as the sum of the real-GDP and deflator log-differences, in the $100 \cdot \ln$ space of Section 2.3:

$$g_t^n \;=\; \left(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\right) \;+\; \left(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\right), \tag{40}$$

where ℓ is the stored $100 \cdot \ln$ level and the right-hand side is already in percent per quarter (it is not annualized by the factor of four used for display in (2)). The primary balance is the revenue–expenditure gap (positive = surplus),

$$pb_t \;=\; rev_t \;-\; primexp_t, \tag{41}$$

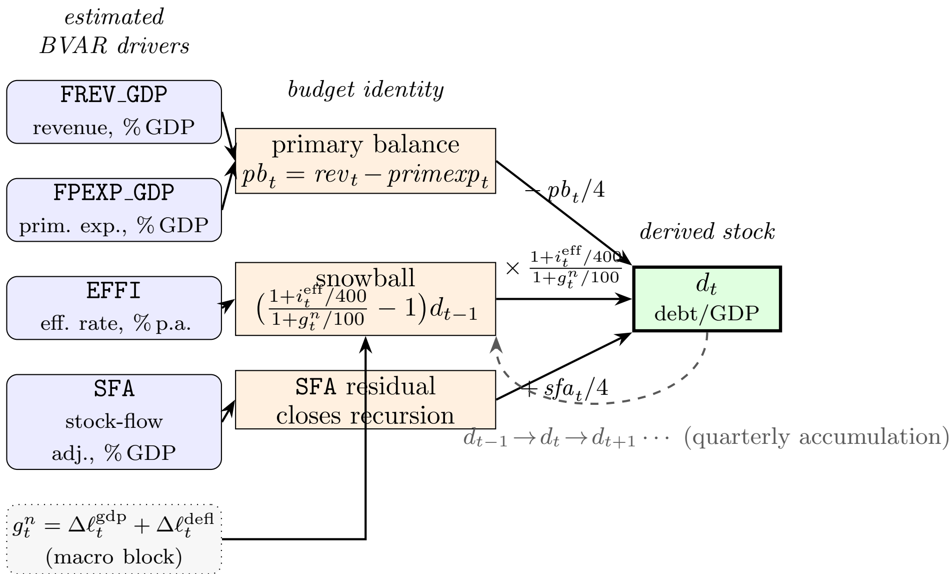

Figure 5: Fiscal stock-flow schematic. The four estimated BVAR drivers (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`; Section 7.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (41); the effective rate `EFFI` and nominal growth g_t^n (40) form the interest–growth *snowball* (46); and the estimated `SFA` enters as the residual that closes the recursion. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1}\,(1+i_t^{\text{eff}}/400)/(1+g_t^n/100) - pb_t/4 + sfa_t/4$ (44), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 7.2).

the interest bill (debt service) is the estimated effective rate applied to the lagged debt stock,

$$int_t \;=\; i_t^{\text{eff}}\,\frac{d_{t-1}}{100}, \tag{42}$$

i.e. the BVAR forecast of the effective rate `EFFI` drives the interest flow directly, with the lagged debt d_{t-1} converting the rate into a percent-of-GDP flow. (A legacy debt-service block would instead supply int_t as the estimated `DSVC` ratio and back out the effective rate as $i_t^{\text{eff}} = 100\,int_t/d_{t-1}$.) The overall balance and deficit follow as

$$ob_t \;=\; pb_t - int_t, \qquad def_t \;=\; -\,ob_t \;=\; int_t - pb_t. \tag{43}$$

The debt stock is then rolled forward one quarter by the FPP/DSA debt-dynamics equation,

$$d_t \;=\; d_{t-1}\,\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100} \;-\; \frac{pb_t}{4} \;+\; \frac{sfa_t}{4}, \qquad d_{-1} \equiv d_0, \tag{44}$$

seeded at d_0. The rate enters quarterized as $i_t^{\text{eff}}/400$ and growth as $g_t^n/100$, while the annual flow ratios pb_t and sfa_t are scaled by 1/4 so that one quarter's worth of the flow accrues per step; sfa_t is the stock-flow adjustment (the deficit–debt discrepancy of Section 7.1). In the current blocs sfa_t is the BVAR's *forecast* of the estimated `SFA` column (read off the level-space path quarter by quarter), so the debt projection carries the discrepancy forward with its own dynamics and bands; absent an estimated SFA column the recursion falls back to a constant default. Gross financing needs add amortization to the deficit,

$$gfn_t \;=\; def_t \;+\; amort_t. \tag{45}$$

The recursion in (41)–(45) is the function `computeFiscalBlock` (with the lever-augmented variant in `runFiscalRecursion`); the exact quarterly convention is pinned by a historical back-cast test that reconstructs observed debt-to-GDP from the realized drivers.

Snowball decomposition. Subtracting d_{t-1} from (44) isolates the *automatic* debt dynamics from the discretionary primary balance. The change in the debt ratio is

$$\Delta d_t \;=\; \underbrace{\left(\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100}-1\right)d_{t-1}}_{\text{snowball (interest–growth)}} \;-\; \frac{pb_t}{4} \;+\; \frac{sfa_t}{4} \;\approx\; \frac{i_t^{\text{eff}}-g_t^n}{1+g_t^n}\,d_{t-1} \;-\; pb_t \;+\; sfa_t, \tag{46}$$

where the right-hand approximation is the familiar Fiscal-Monitor $(i-g)$ form (here in annualized flow units). The snowball term is positive—debt rises mechanically—whenever the effective rate exceeds nominal growth; a primary surplus and a favourable growth-interest differential are the two forces that stabilize the ratio. The tool returns the per-quarter snowball contribution $\big((1+i_t^{\text{eff}}/400)/(1+g_t^n/100)-1\big)d_{t-1}$ alongside the derived balances so the user can read off the source of any debt drift.

Remark 2. Because the debt stock is *not* a BVAR variable, the estimated effective rate $\hat{\imath}_t^{\text{eff}}$ and primary balance $\widehat{pb}_t$ are debt-agnostic baselines: the recursion uses *lagged* debt d_{t-1} throughout, so it stays explicit (no within-period fixed point) and the active levers of Section 7.3 layer on without double-counting. This is the design that lets a single forward sweep produce a consistent debt path.

7.3 Active Levers

The accounting layer exposes three user-editable levers with literature-based defaults, governed by a master toggle (`feedbackOn`, default on). They modify the estimated baselines in terms of the lagged debt gap $d_{t-1}-d^*$, where the anchor d^* defaults to d_0.

Bohn fiscal-reaction function (γ). Following Bohn (1998), the primary balance is allowed to respond to the lagged debt gap, so that fiscal policy consolidates as debt rises above its anchor:

$$pb_t \;=\; \big(rev_t - primexp_t\big) \;+\; \gamma\,\big(d_{t-1}-d^*\big), \qquad \gamma \geq 0. \tag{47}$$

The default $\gamma = 0.03$ (three basis points of primary surplus per percentage point of excess debt) is the canonical Bohn (1998) estimate and the threshold above which a no-Ponzi debt process is mean-reverting. Setting $\gamma = 0$ disables the reaction.

Risk-premium channel (β). The effective rate is allowed to rise with debt above the anchor, capturing a sovereign risk premium:

$$i_t^{\text{eff}} \;=\; \hat{\imath}_t^{\text{eff}} \;+\; \beta\,\max\!\big(0,\; d_{t-1}-d^*\big), \qquad \beta \geq 0, \tag{48}$$

where the premium bites only *above* the anchor (the $\max(0,\cdot)$ rectifier). The default is $\beta = 0$: the magnitude of the debt-premium elasticity is empirically contested, so the lever is shipped inactive and left to the user. When active, β feeds back into both the interest bill (42) and the snowball term (46), producing the nonlinear debt-spiral dynamics that motivate the FPP/DSA realism literature.

Stock-flow-adjustment closure. The SFA term sfa_t in (44) is now *estimated* as its own BVAR variable (Section 7.1), so the default debt projection already carries the forecast SFA path forward with dynamics and bands rather than defaulting the discrepancy to zero. It can still be overridden—set to zero for pure flow accounting, held at the historical average, or used as a scenario lever to reconcile a target debt path with the flow drivers.

Both feedback channels act on the lagged gap and through the lagged debt already entering (42)–(44), so they are added cleanly to the debt-agnostic baselines without re-estimation and without double-counting.

7.4 Debt-Targeting Inversion

The recursion can also be run backwards. Given a desired debt-to-GDP path $\{d_t^{\text{target}}\}$, the effective rate, nominal growth, and SFA, the annual primary balance required at each quarter is obtained by solving (44) for pb_t along the target trajectory:

$$pb_t \;=\; 4\left[d_{t-1}^{\text{target}}\,\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100} \;-\; d_t^{\text{target}}\right] \;+\; sfa_t, \tag{49}$$

with $d_{-1}^{\text{target}} \;\equiv\; d_0$. This is the IMF DDT/DSA "required primary balance / adjustment to the debt anchor" (`requiredPrimaryBalance`). The caller then conditions the BVAR on this primary-balance path through the linear constraint $rev_t - primexp_t = pb_t$, propagated by the DK smoother of Section 4. Because the effective rate and growth themselves respond to the primary balance through the BVAR covariance structure, an exact hit is obtained by iterating invert $\rightarrow$ condition $\rightarrow$ re-read $(i, g) \rightarrow$ re-invert; one or two passes converge.

7.5 Display Layout

The estimated and derived fiscal objects are presented in different chart rows, reflecting their different epistemic status. The four *estimated* fiscal drivers are shown as inputs alongside the macro block: revenue and primary expenditure as % of GDP, and the two new charts

- **Effective interest rate** (`EFFI`, % p.a.): the average rate the government pays on its outstanding stock. It is plotted as a level in percent per annum next to the policy rate and the market yields. Read it as the slow-moving cost of the debt: it lags the policy rate and the 10-year yield, because only maturing securities reprice, so a rate hike shows up here gradually rather than at once. This is the rate that enters the interest bill (42) and the snowball (46).

- **Stock-flow adjustment** (`SFA`, % of GDP): the deficit–debt discrepancy. It is plotted as a level in percent of GDP. A positive value means gross debt rose by more than the deficit plus interest–growth dynamics imply (debt-increasing other-means-of-financing operations); a value near zero means the flow accounting closes cleanly. Its history is informative—it is large and persistent for the high-debt blocs (Table values of order +2 to +4% of GDP for JP, UK, CA, CN)—and, crucially, its *forecast* now feeds the debt projection, so a persistently positive SFA pushes the projected debt path *above* what the deficit alone would deliver.

A further row holds the *derived* reporting variables produced by the accounting recursion, in the order

(i) **Primary Balance** (pb_t, % of GDP), which is itself conditionable: the user can drag a primary-balance path and the tool maps it back to the $rev - primexp$ linear combination;

(ii) **Δ Debt-to-GDP** (Δd_t, percentage points per quarter), the per-quarter debt change of (46), which is not directly conditionable because it is a pure identity output;

(iii) **Debt-to-GDP** (d_t, % of GDP), the rolled-forward stock, which can be targeted via the inversion of Section 7.4. Note that this projection now *carries the forecast SFA forward* (with dynamics and bands): a bloc with a large positive estimated SFA will show a debt path that drifts up faster than its primary balance and snowball alone explain, which is the intended, honest behaviour.

Separating the estimated drivers (effective rate, SFA, revenue, expenditure) from the derived balances and debt makes the stock-flow-consistent structure visible on screen: what the data and the prior speak to is shown as an input, and what the budget identity implies is shown as an output.

7.6 Data Treatment: the U.K. Index-Linked-Gilt Spike

Interest series constructed from accrued interest can contain mechanical, non-recurring spikes that are not informative about the underlying refinancing cost. In the United Kingdom, the large stock of *index-linked gilts* produced an interest spike in 2008–09: the inflation uplift on the principal of index-linked debt is recorded as accrued interest, so the 2008 inflation spike (and its subsequent unwinding) translated into an outsized swing in the measured effective rate `EFFI` that would otherwise contaminate the BVAR's autoregressive dynamics. These quarters are handled with the same machinery used for the COVID-19 observations: the affected effective-rate entries are set to missing (`NaN`) and imputed by the Kalman filter inside the data-augmentation Gibbs sampler, rather than being allowed to enter the likelihood as if they were ordinary observations. This keeps the estimated effective-rate process—and hence the derived interest bill and debt path—free of a one-off accounting artefact while preserving the genuine information in the surrounding quarters.

8 Long-Run Anchoring: Prior vs Entropic Tilt

The forecasts of Section 4 inherit their long-run behaviour entirely from the estimated VAR. Because the real-activity and price blocks are estimated on $100 \ln$ levels of $I(1)$ series, the companion matrix (6) carries roots on or near the unit circle, and the implied long-horizon central path can drift away from any value the user regards as a credible steady state (e.g. a 2% inflation objective, a neutral policy rate, or a balanced great ratio). The tool exposes a *long-run anchor* control that lets the user pin the long-run forecast of a chosen linear combination $h'y$ to a target. We support two conceptually distinct mechanisms, summarized in Table 2: an *in-prior* long-run prior applied *before* estimation, and an *after-estimation* entropic tilt of the predictive density. They answer different questions, and the distinction is methodologically important.

8.1 In-Prior Long-Run Prior (PLR)

The first mechanism is the "prior for the long run" (PLR) of Giannone et al. (2019), which augments the hierarchical Minnesota prior of Section 2.4 with additional dummy observations that shrink a small set of linear combinations Hy toward stationarity (no-cointegration / random-walk behaviour). Let $H \in \mathbb{R}^{q\times n}$ collect q long-run combinations and let y_0 be the prior pre-sample mean. Following the authors' construction, H is rotated into a full-rank basis $\bar{H} = \big(H';\, \mathrm{null}(H)'\big)'$ with inverse $\bar{H}^{-1}$, and for each active long-run row j a single dummy observation

$$y_j^d = \frac{(y_0\bar{H}')_j}{\phi_j}\,\big(\bar{H}^{-1}\big)'_{\cdot j}, \qquad x_j^d = \Big(0,\ \tfrac{(y_0\bar{H}')_j}{\phi_j}\,\mathbf{1}_p \otimes \big(\bar{H}^{-1}\big)'_{\cdot j}\Big), \tag{50}$$

is appended *to the regression system* (both the Gibbs draw and the marginal-likelihood normalizing constant T_d), with ϕ_j controlling the tightness of the j-th long-run restriction. Crucially, (50) combines the *equations* through the columns of $\bar{H}^{-1}$; combining the regressors instead produces a dummy whose residual is identically zero at the sum-of-coefficients mode and hence carries no tightness information. The tightness hyperparameters ϕ_j are held fixed so that the Metropolis–Hastings search over the Minnesota λ and the sum-of-coefficients μ remains two-dimensional.

The PLR is a *proper Bayesian prior*: it modifies the posterior and requires re-estimation. It is therefore computed offline as part of the data-export pipeline, not in the browser.

Remark 3 (What the PLR actually does in the US sample). A US head-to-head ($n = 36$, $p = 3$, COVID treated as missing per Section 4.6, $q = 13$ long-run combinations comprising six great ratios, three nominal anchors, and four rate/credit spreads) shows that the PLR is a *separate dummy block* rather than a change to the Minnesota tightness: the maximizing λ is essentially unchanged (0.563 baseline vs 0.562 with the PLR at either $\phi = 1.0$ or $\phi = 0.1$). Because the sum-of-coefficients-style dummy in (50) shrinks toward a *random walk* (no cointegration), and the US sample treats most of these ratios as genuinely $I(1)$, the PLR does *not* pull the long-run central forecast toward a historical mean: the mean absolute drift of the $H{=}40$ forecast from a recent historical anchor moves from 8.49 (baseline) to 8.41 at $\phi = 1.0$ (-1%), and is marginally *worse* at $\phi = 0.1$. The visible effect is on the *bands*, not the path: the GDP-growth 68% band tightens from 4.24 to 3.95 percentage points ($\approx 7\%$) at $\phi = 1.0$. The PLR is thus the appropriate instrument for principled band tightening and a soft no-cointegration restriction; it is *not* a device for pinning a long-run combination to a chosen number.

8.2 After-Estimation Entropic Tilt

The second mechanism is the user-facing one, exposed in the interface as the **Long-run priors** panel (Section 10). It does *not* re-estimate the model. Instead it takes the *existing* posterior draws $\{(\hat\beta^{(d)}, \hat\Sigma^{(d)})\}_{d=1}^{D}$ of Section 4.6 and applies an exponential (entropic) tilt — a relative-entropy reweighting in the spirit of Robertson et al. (2005) — so that the *weighted* mean of each long-run moment matches the user's target. The result is a vector of per-draw importance weights that the band loop applies through weighted quantiles; the underlying draws are never re-solved.

The long-run moment. For draw d the relevant long-run quantity is taken to be the *long-horizon mean forecast*, not the algebraic steady state:

$$g^{(d)} = h' \, \mathbb{E}\big[y_{T+H} \mid \hat\beta^{(d)}\big], \tag{51}$$

obtained by iterating the deterministic companion recursion (5) forward H steps with the innovations switched off, exactly as in the mean iteration underlying (21). The default horizon is $H = 40$ (≈ 10 years quarterly), matching the convention of Giannone et al. (2019). We deliberately do *not* use the closed-form steady state $y^\star = (I - \sum_\ell B_\ell)^{-1} c$, $g = h' y^\star$. For the near-unit-root level VARs the matrix $(I - \sum_\ell B_\ell)$ is near-singular, so $y^\star$ is a meaningless quantity of order 10^5: e.g. for the US GDPH $-$ CH log spread the steady-state evaluation returns $\approx -2.1 \times 10^4$, whereas the $H{=}40$ forecast moment is the finite $\approx +35$ that matches the chart. The steady-state evaluation is retained only as an opt-in for genuinely stationary combinations.

For a single user variable the moment is stated in the chart's own display units (Section 2.3): for a `lin` variable (rates, unemployment) the target is the forecast *level* $e_j' \mathbb{E}[y_{T+H}]$; for a `log` variable it is the annualized growth $4(\mathbb{E}[y_{T+H,j}] - \mathbb{E}[y_{T+H-1,j}])$, consistent with (20), so the user enters the number they read off the axis.

KL-minimizing weights. Collecting q restrictions with targets $\bar g_r$ and tightness scales s_r, define the centered, scaled moments $\tilde g_r^{(d)} = (g_r^{(d)} - \bar g_r)/s_r$. We seek weights $w_d \propto \exp\big(\gamma' \tilde g^{(d)}\big)$ that satisfy the moment conditions $\sum_d w_d \, g_r^{(d)} = \bar g_r$ for all r while minimizing the Kullback–Leibler divergence (Kullback and Leibler, 1951) from the uniform weights $1/D$. This is the classical exponential-tilting / minimum-relative-entropy projection: the tilted measure is the I-projection of the empirical predictive measure onto the affine set of distributions satisfying the long-run moment constraints. The Lagrange dual is the smooth, convex objective

$$F(\gamma) = \log\left(\frac{1}{D}\sum_{d=1}^{D}\exp\!\big(\gamma'\,\tilde{g}^{(d)}\big)\right), \qquad \nabla F(\gamma) = \mathbb{E}_w\big[\tilde{g}\big], \qquad \nabla^2 F(\gamma) = \mathrm{Cov}_w\big(\tilde{g}\big), \tag{52}$$

whose stationary point $\nabla F(\gamma^\star) = 0$ enforces the constraints. We solve (52) by Newton iteration with a backtracking (Armijo) line search and a 10^{-10} ridge on the Hessian, computing F via a numerically stable log-mean-exp; the weights match the target mean exactly regardless of the scales s_r, which serve only to balance restrictions of different magnitudes (each set to the moment's own posterior standard deviation). The normalized weights w_d are applied as a weighted reduction of the draw cloud: they move the *unconditional central path* (the displayed baseline becomes the weighted mean of the draws, not the unweighted posterior-mean path) and the band quantiles together, on *both* the baseline fan (the precomputed band of Section 4.6 is replaced by the in-browser tilted band) and the conditional/scenario fan, so the entire predictive density—central path included—is pulled toward the anchor.

Effective sample size and degeneracy. The tilt is a pseudo-posterior reweighting, and its reliability is governed by the effective sample size

$$\mathrm{ESS} = \frac{\big(\sum_d w_d\big)^2}{\sum_d w_d^2} = \frac{1}{\sum_d w_d^2} \in [1, D], \tag{53}$$

the harmonic measure of how many draws carry non-negligible mass after reweighting. As the target is pushed farther from the prior mean $D^{-1}\sum_d g^{(d)}$, the weights concentrate and ESS falls. The interface surfaces ESS$/D$ as a color-coded badge (green above 40%, amber between 10% and 40%, red below 10%) so the user can see when an anchor is straining the support of the draws.

Remark 4 (Per-constraint vs joint tilt). In the US comparison of Remark 3, with $D = 500$ draws a *per-constraint* tilt — anchoring one combination at a time — hits every target exactly with a healthy effective sample size (median ESS $\approx$ 75.9% of D). The *joint* 13-constraint tilt, by contrast, is degenerate: ESS $\to 1$, because the joint anchor target lies outside the 13-dimensional convex hull of only 500 draws even though each marginal target lies inside its own one-dimensional hull. The practical consequence, reflected in the interface, is that the entropic tilt is well-behaved for one or a few long-run restrictions but should not be over-constrained: the panel defaults to a single anchor and warns when several are enabled simultaneously.

8.3 When to Use Which

The two mechanisms are complementary. The in-prior PLR is the gold standard — a proper offline posterior — but its data-decides property means that, at the tightnesses examined, it leaves the US central long-run forecast essentially unchanged and acts mainly on the bands. The entropic tilt is in-browser, requires no re-estimation, and *forces* the weighted long-run forecast onto an explicit target, but it is a reweighting of a finite draw cloud and degrades (falling ESS) if over-constrained. As a rule of thumb: to *pin* a long-run combination to a number, use the entropic tilt (or, offline, a tighter targeted prior); for principled band tightening and a soft no-cointegration restriction, use the PLR.

Table 2: Long-run anchoring: in-prior PLR versus after-estimation entropic tilt.

	In-prior PLR	**Entropic tilt**
Source	Giannone et al. (2019)	Robertson et al. (2005)
Acts on	prior (dummy obs.)	posterior draws (weights)
Re-estimation	yes (offline)	no (in-browser)
Object	proper posterior	pseudo-posterior
Moment	Hy stationarity	$h'\mathbb{E}[y_{T+H}]$
Main effect	tightens bands	shifts central path
Failure mode	none (proper)	low ESS if over-constrained

9 Economy Coverage, Data Sources, and Estimation

The tool ships estimated BVARs for roughly fifty economies. Thirty-three of them are integrated into a single *Global VAR* (GVAR) system (Pesaran et al., 2004; Dees et al., 2007) covering approximately 90% of world GDP; the remaining blocs—including the 17 euro-area member states, which are deliberately kept out of the coupled system because a disaggregated euro GVAR is non-stationary—are standalone single-country models that share the same estimation machinery but are not wired into the cross-country fixed point. This section documents the roster, the underlying data, the COVID-as-missing estimator, the common-vintage harmonization, and the construction of the predictive bands. Throughout, variables enter the VAR in the $100 \ln(\cdot)$ space for `log` series and in levels for `lin` series (Section 2.3); annualized quarter-over-quarter growth and inflation are reported as $4 \times \Delta(100 \ln)$, consistent with (2) and (20).

9.1 The GVAR System: 33 Integrated Blocs

The GVAR comprises the United States—now a fully endogenous coupled bloc, not a frozen anchor (Section 6.1)—the euro area as a single aggregate bloc, plus 31 further Rest-of-World (RoW) blocs:

> US; EA (euro-area aggregate), JP, UK, CA, CN (the core majors); CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, TH, MY; and the nine further non-euro economies CL, CO, CR, CZ, HU, IL, IS, NZ, TW added in the OECD expansion.

The euro area enters as one aggregate bloc by necessity: a fully disaggregated euro GVAR is non-convergent ($\rho(M) \approx 1.089 > 1$), and treating the euro area as a single bloc restores stationarity.

The 17 euro-area members are consequently *standalone* models (Section 9.2), not coupled GVAR blocs. Each bloc—the US included—is an independently estimated single-country BVAR in which the foreign sector is summarized not by a raw foreign-variable block but by three trade-weighted RoW "star" variables, following the GVAR tradition of Pesaran et al. (2004) and Dees et al. (2007):

- `ROWDEM` — foreign demand: the trade-weighted average of partners' real-GDP growth;
- `ROWINF` — foreign inflation: the trade-weighted average of partners' price inflation;
- `ROWFIN` — foreign financial conditions: the trade-weighted average of partners' short rates.

For bloc c with partner set $\mathcal{P}_c$ and weights w_{cj}, the demand and inflation stars are built in annualized growth space from the partners' stored $100\ln$ levels,

$$\mathrm{ROWDEM}_{c,t} = 4 \times \frac{\sum_{j\in\mathcal{P}_c} w_{cj}\left(x^g_{j,t} - x^g_{j,t-1}\right)}{\sum_{j\in\mathcal{P}_c} w_{cj}\,\mathbb{1}\{x^g_{j,t}, x^g_{j,t-1}\ \text{observed}\}}, \tag{54}$$

where $x^g_{j,t}$ is partner j's real-activity level (analogously for `ROWINF` from prices, and `ROWFIN` from the level of the short rate). Weights are *renormalized per component* over the partners actually observed in each quarter, so a partner that is missing a given series simply drops out of the corresponding star and the remaining weights are rescaled. Two implementation details matter for the econometrician: China's reported GDP is nominal, so its real-activity level is formed by subtracting the price level in $100\ln$ space, $x^g_{\mathrm{CN}} = x^{\mathrm{NGDP}}_{\mathrm{CN}} - x^P_{\mathrm{CN}}$ (a log-space deflation); and Argentina's 2023–24 hyperinflation episode is excluded from the `ROWINF` and `ROWFIN` aggregates of its partners (its genuine real demand is retained in `ROWDEM`), with renormalization absorbing the dropped weight.

Trade weights. Bilateral weights w_{cj} are constructed from goods-trade flows in the IMF International Trade in Goods statistics (IMTS, the renamed bilateral DOTS), queried from the SDMX 2.1 endpoint `api.imf.org` over the 2021–2023 average of exports (`XG_FOB_USD`) plus imports (`MG_CIF_USD`). The Euro Area is treated as a single counterpart (IMTS aggregate code `G163`). The weight matrix is row-normalized over the 33 blocs with a zero diagonal; now that the US is endogenous it carries its own trade-weights row, so all 33 blocs have a row.

Rate-optional blocs. The short rate is *optional*: a GDP series and a price series are required of every bloc, but two blocs (Switzerland and Norway) carry no domestic short rate. Their rate column is `NaN` throughout, they are excluded from their partners' `ROWFIN` aggregate, and the `ROWFIN` weights of any bloc having them as a partner are renormalized accordingly. They still contribute to `ROWDEM` and `ROWINF`.

Cross-country coupling. As in Section 9's original treatment, the baseline forecast of each bloc is computed independently; cross-country coupling enters only through the scenario layer. A shock conditioned in one bloc perturbs its star variables, which propagate to its trade partners, whose own stars then feed back. Stacking the per-bloc deviations Δ yields the linear fixed point

$$\Delta = M\,\Delta + d, \tag{55}$$

where M encodes the trade-weighted star couplings together with the four US-determined global commons (the spot oil price, the US 10-year yield, the US Baa corporate yield, and US equity; Section 6.6) and d collects the user's direct conditioning forcing. With the US endogenous the deviation state is *augmented* to carry the US global-column deviations as unknowns (Section 6.4), so the global commons close the RoW→US feedback loop inside M rather than entering d as a frozen forcing. The system is solved at display time both by a direct $(I - M)^{-1}d$ evaluation and by Neumann sweeps (summing $\sum_k M^k d$); the two agree to $\sim 10^{-9}$. The coupling is contractive, with spectral radius $\rho(M) \approx 0.924$, so the series converges. Because the full 2060×2060 $(I - M)^{-1}$ inverse is expensive in-browser (a cold build is on the order of three-quarters of an hour), it is precomputed offline and shipped as a *gzipped* coupling cache (`coupling_default.json.gz`, about 30 MB compressed), decompressed in the browser via the native `DecompressionStream` API. The interactive interface uses a two-tier solve—a fast direct preview on drag and an explicit "Solve full GVAR" button for the complete fixed point—to keep the endogenous-US solve reactive (Section 6.5).

9.2 Standalone Single-Country Blocs

Beyond the 33 GVAR blocs, the tool ships additional single-country BVARs estimated by the same procedure but not coupled into (55). Together with the GVAR members these complete coverage of all 38 OECD members and add several large non-OECD emerging markets. The standalone remainder is dominated by the 17 *euro-area member states* (DE, FR, IT, ES, NL, AT, BE, FI, GR, IE, LU, PT, SI, SK, EE, LT, LV), which are kept out of the coupled GVAR *by construction*: the euro area is represented in the GVAR by the aggregate `EA` bloc, because a disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$) whereas aggregating the euro area restores convergence. Each euro member is nonetheless estimated as a full single-country model—its domestic block augmented by the four global commons and by trade-weighted foreign *star* variables—but in *separation* (with the GVAR iteration count fixed to `MAXIT` $= 1$, so its foreign stars are read off data rather than from a cross-country solve) and it carries no trade-weights row. The country selector groups the roster into two sections, a *GVAR* group and an *Other countries* group; a bloc is GVAR-linked, and hence placed in the *GVAR* group, if and only if it has a trade-weights row and its specification carries the three `ROW*` star variables.

9.3 Data Sources

No single provider supplies current quarterly real GDP for the full roster, so the data are assembled bloc-by-bloc:

- **FRED.** The US block ($n = 41$, Table 3) and many other series are pulled from the St. Louis Fed's FRED. The China bloc is built entirely from verified real FRED series, with quarterly real activity anchored by industrial production (`PRCNTO01CNQ661S`) since clean quarterly *real* GDP is not available on FRED for China (only annual constant-price GDP and nominal quarterly GDP that ends early).

- **Eurostat.** The IMF's new thematic SDMX API exposes no per-country quarterly chain-volume GDP (its national-accounts flow carries current-price values only, for a limited country set), so the 22 EU/EFTA OECD economies are sourced from Eurostat (`namq_10_gdp` chain-linked volumes, `prc_hicp_midx` HICP, `une_rt_q` unemployment, `irt_*` interest rates, `ert_eff_ic_q` NEER), which is current to 2025Q4–2026Q1.
- **National banks and OECD QNA.** The non-EU OECD members (e.g. Chile, Korea, Australia, New Zealand, Mexico, Türkiye) draw real GDP and national accounts from the OECD Quarterly National Accounts, short rates from national-bank series (e.g. RBA/RBNZ), and CPI, unemployment, and NEER from the IMF flows.

Real versus nominal GDP is handled per bloc: where a chain-volume real series exists it is used directly; China's nominal GDP is deflated in log space as above; and thin or volatile samples are estimated with deliberately wide, honest predictive bands rather than spuriously tight ones.

9.4 COVID-as-Missing Estimation

The 2020 pandemic quarters are an extreme outlier that, left untreated, would contaminate the autoregressive estimates and the innovation covariance. Rather than scaling them out, the tool treats the pandemic window as *missing data* and imputes it within the estimation itself, via a data-augmentation Gibbs sampler that nests a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012). Each Gibbs sweep alternates:

(i) a Durbin–Koopman simulation-smoother draw of the missing pandemic rows given the current coefficients, treating those rows as `NaN` observations in the companion-form state space of Section 3;

(ii) a Metropolis–Hastings update of the hierarchical-prior hyperparameters on the *completed-data* marginal likelihood, so the overall tightness λ and the sum-of-coefficients parameter μ (Giannone et al., 2015) are selected by empirical Bayes without any fixed-tightness restriction; and

(iii) a draw of the VAR coefficients (β, Σ) from the completed data.

The pandemic history is thus *preserved* (the surrounding quarters remain in the sample and inform the imputation) while the outlier itself does not drive the dynamics.

Window selection. The COVID window self-calibrates to each bloc's pandemic trough rather than being hard-coded. It is the *argmin* of the quarter-on-quarter GDP change *restricted to the*

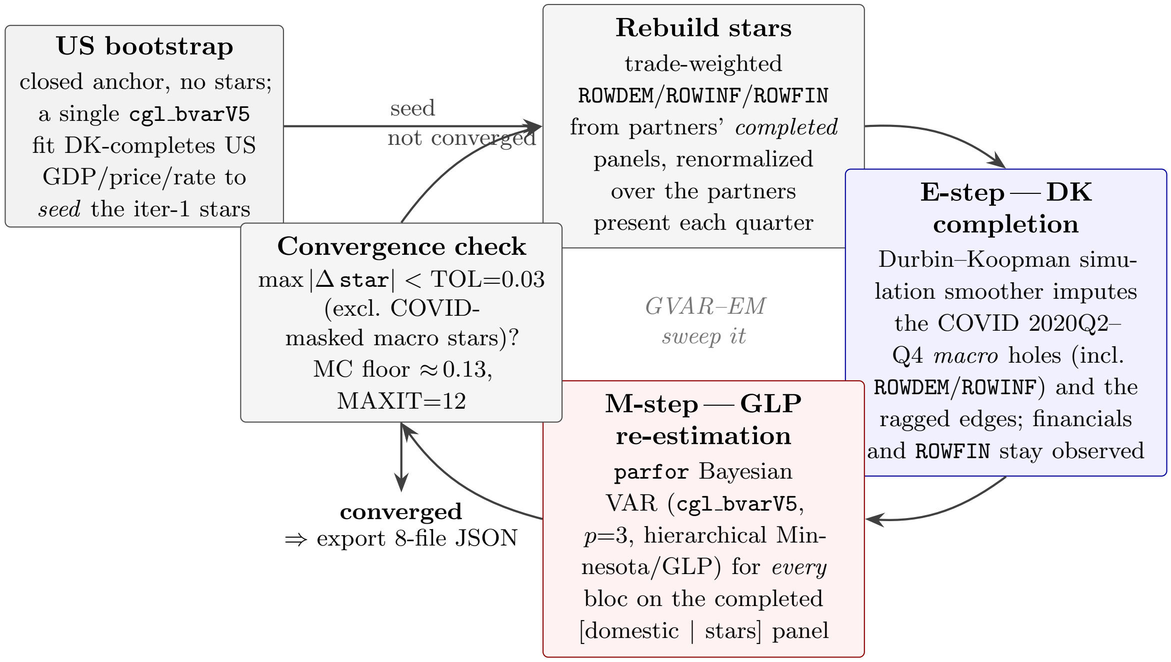

Figure 6: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the `ROWDEM/ROWINF/ROWFIN` stars from partners' completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc's single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and `ROWFIN` held observed) and the ragged, unbalanced-panel edges—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, max $|\Delta\, \texttt{star}| <$ TOL $= 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is Monte-Carlo noise with floor ≈ 0.13), capped at MAXIT $= 12$ iterations.

window 2019Q4–2021Q1:

$$k^\star = \underset{t\,\in\,\{2019\text{Q4},\ldots,2021\text{Q1}\}}{\arg\min} \left(x_t^g - x_{t-1}^g\right), \qquad \mathcal{W} = \{\, k^\star - 1,\; k^\star,\; k^\star + 1,\; k^\star + 2 \,\}, \tag{56}$$

and the four rows of $\mathcal{W}$ (intersected with the sample) are NaN'd across *all* variables before estimation. Restricting the search to 2019Q4–2021Q1 is deliberate: an unrestricted global argmin misfires for economies whose deepest contraction is a different crisis (for example Korea's 2008Q4 Global-Financial-Crisis drop exceeds its 2020Q2 pandemic drop).

Fixed window and macro-only masking in the production GVAR loop. The auto-detected window above is the legacy single-bloc convention; the production GVAR estimator (`pipeline/export_bvar_gvar24` instead uses a *fixed* three-quarter window $\mathcal{W} = \{2020\text{Q2},\ 2020\text{Q3},\ 2020\text{Q4}\}$ for every bloc, which avoids the ∼seven-quarter spans the unrestricted trough search could produce. Equally important, the production loop masks *only the macro columns*: over $\mathcal{W}$ the domestic macro variables (`isFinancial` $=$ false) and the two macro stars `ROWDEM`/`ROWINF` are set `NaN`, while the domestic financial series (rates, yields, equity, house prices, corporate spreads) and the financial star `ROWFIN` are *kept observed*. The Durbin–Koopman smoother then imputes the pandemic macro holes *conditional on* the observed financials, which carry genuine signal through 2020—the masking is thus "macro-masked / financials-observed," the same convention the out-of-sample evaluation re-applies in-sample at every origin (Section 13.2).

Algorithm 3 states the full estimation loop. It is the operational form of the cycle sketched in Figure 6: a no-stars US bootstrap seeds the first foreign variables, after which each sweep rebuilds every bloc's trade-weighted stars from partners' completed panels (the *E-step* regressor), re-estimates all blocs in parallel under macro masking (the per-bloc `cgl_bvarV5` call, whose own Durbin–Koopman completion is the within-bloc *E-step* and whose Gibbs re-estimation is the *M-step*), and recomputes the stars from the freshly completed data, iterating to a fixed point in the completed stars.

Algorithm 3 EM-style GVAR estimation loop (Jacobi star re-convergence; `export_bvar_gvar24_loop.m`).

Input: per-bloc panels $\{X_c\}$ ($X_c = [\text{domestic} \mid \text{stars}]$); trade weights W; lags $p = 3$; `WN`/stationary index sets $\texttt{pos}_c$; tolerance $\text{TOL} = 0.03$; cap $\text{MAXIT} = 12$; fixed COVID window $\mathcal{W} = \{2020\text{Q}2..\text{Q}4\}$.
Output: per-bloc posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$, completed panels, predictive draws $\texttt{Dforecast}_c$, exported 8-file JSON.

▷ Bootstrap (seed the first stars).
Mask the US macro columns over $\mathcal{W}$ and fit the closed, no-stars US bloc with `cgl_bvarV5`; the posterior-mean DK completion gives the initial US real-GDP/price/rate panel (this seeds partners' iteration-1 stars).
for $it = 1, \ldots, \text{MAXIT}$ **do** ▷ Jacobi sweep
  ▷ Build all stars from the FROZEN start-of-sweep completed panels.
  for each re-estimated bloc c: $\texttt{star}_c \leftarrow$ trade-weighted `ROWDEM`/`ROWINF`/`ROWFIN` from partners' completed panels (renormalize over partners present each quarter).
  parfor each bloc c **do** ▷ independent given the frozen stars
    $x_c \leftarrow [\,\text{domestic}_c \mid \texttt{star}_c\,]$
    over $\mathcal{W}$, set `NaN` the macro columns of x_c *and* `ROWDEM`,`ROWINF` (keep financials, `ROWFIN`)
    $(\texttt{Xcond}_c, \beta_c, \Sigma_c, \texttt{Dforecast}_c) \leftarrow \texttt{cgl\_bvarV5}(x_c, p, \texttt{pos}_c, \ldots)$ ▷ E-step DK completion + M-step Gibbs
    completed panel $\widehat{X}_c \leftarrow \text{mean}_{\text{draws}}\, \texttt{Xcond}_c$
  end parfor
  $mc \leftarrow \max_c \max \left| \texttt{star}_c^{(it)} - \texttt{star}_c^{(it-1)} \right|$ ▷ exclude COVID-masked `ROWDEM`/`ROWINF` cells
  if $it > 1$ **and** $mc < \text{TOL}$ **then break** ▷ converged
end for
Export each bloc's $(\hat{\beta}_c, \hat{\Sigma}_c)$, completed history, stars, and predictive draws to the 8-file JSON contract.

The sweep is a *Jacobi* update—all stars are rebuilt from the common start-of-sweep state *before* the parallel re-estimation—so the parallel blocs are mutually independent given the current stars and the sweep has the same fixed point as a sequential (Gauss–Seidel) pass, at a wall-clock set by the slowest single bloc (the $n = 41$ US bloc) rather than the sum. The convergence metric excludes the COVID-masked macro-star cells, whose cross-sweep churn is pure imputation Monte-Carlo noise (floor ≈ 0.13); the financial star `ROWFIN`, kept observed through the pandemic, stays in the metric. Each `cgl_bvarV5` call is itself the data-augmentation Gibbs sampler of Algorithm 1—its DK completion is the bloc-level E-step and its hyperparameter/coefficient draws are the M-step—so the whole loop is a block-coordinate EM over (foreign stars, completed-data states, VAR coefficients).

9.5 Common-Vintage Harmonization to 2025Q4

So that the first forecast (nowcast) quarter is uniform across the tool, every bloc's domestic series is extended to a common 2025Q4 endpoint, making 2026Q1 the first projected quarter everywhere. For blocs whose quarterly data end short of 2025Q4:

- **Real-activity and price series** (GDP, consumption, investment, government, exports, imports, CPI/HICP, all `log`) are benchmarked to IMF World Economic Outlook annual projections (`NGDP_RPCH` real growth, `PCPIPCH` inflation) via temporal disaggregation: a within-year quarterly profile is laid down at the WEO-implied annual growth and then proportional-Denton corrected so that each fully appended year's four-quarter average reproduces the WEO annual level (already-observed quarters in a partial year are held fixed). The realized annual growth of each appended year matches WEO to within 0.2pp.
- **Financial series with no WEO analogue** (NEER, REER, equity indices, and all `lin` rates — short rates, 3-month rates, 10-year yields, unemployment) are carried forward by *hold-last* and flagged as such.

The original raw vintages are left untouched; the extended panels are written separately and consumed by the estimation drivers.

9.6 Predictive Uncertainty Bands

For every bloc the exported baseline carries *predictive* bands, not parameter-only bands. The COVID-missing Gibbs sampler emits a forecast density that integrates over three sources of uncertainty simultaneously — posterior parameter uncertainty, future innovation uncertainty, and missing-data (imputation) uncertainty. The reported quantiles are the 5th, 16th, 84th, and 95th percentiles of this predictive density (with the 50th retained in level space),

$$\left(q_{05},\, q_{16},\, q_{84},\, q_{95}\right)_{T+h} = \mathrm{Quantile}_{\{0.05,0.16,0.84,0.95\}}\left\{\, \tilde{y}^{(d)}_{T+h} \,\right\}_{d=1}^{D}, \tag{57}$$

yielding the 68% band (q_{16}, q_{84}) shaded in the charts and the 90% band (q_{05}, q_{95}). Crucially, the central path $\bar{y}^{\text{baseline}}$ remains the *deterministic* forecast from the posterior-mean coefficients (so the unconditional smoother of Section 4.5 reproduces it exactly, and zero user deviations return the stored baseline); only the bands are predictive. This is the designed fix for the previously over-tight bands that resulted from spreading posterior-mean paths under a tight prior — wide bands now reflect genuine macro volatility (e.g. CN, JP, UK), not the COVID outlier. On-the-fly conditional bands computed in the browser use the same predictive construction, drawing innovations through the simulation smoother of Section 4.6.

9.7 US Specification

The US is the largest bloc, with $n = 41$ quarterly variables and $p = 3$ lags. Its full variable list is given in Table 3; the four-variable fiscal block (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`) supports the fiscal-reaction counterfactuals in the spirit of Bohn (1998), and the conditional-forecast and optimal-policy machinery of Bańbura et al. (2015) and Barnichon and Mesters (2023) applies uniformly across all blocs. The count of 41 comprises the domestic and fiscal variables, the four global commons, and the three trade-weighted `ROW*_US` star variables that make the US a fully coupled GVAR bloc (Section 6.1).

Table 3: Variable coverage for the US BVAR model. "Trans." denotes the data transformation: `log` = log levels (growth rates displayed), `lin` = levels. "Prior" denotes the prior mean: RW = random walk, WN = white noise.

ID	**Description**	**Trans.**	**Prior**
GDPH	Real Gross Domestic Product	log	RW
CH	Real Personal Consumption Expenditures	log	RW
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW
XH	Real Exports of Goods & Services	log	RW
MH	Real Imports of Goods & Services	log	RW
GH	Real Government Consumption Expenditures & Gross Investment	log	RW
JGDP	Gross Domestic Product: Chain Price Index	log	RW
JC	Personal Consumption Expenditures: Chain Price Index	log	RW
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW
UI	CPI-U: All Items	log	RW
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW

ID	Description	Trans.	Prior
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW
LANAGRA	All Employees: Total Nonfarm	log	RW
LR	Civilian Unemployment Rate: 16 yr +	lin	RW
IP	Industrial Production Index	log	RW
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW
HST	Housing Starts	log	RW
YPDH	Real Disposable Personal Income	log	RW
CSENT	University of Michigan: Consumer Sentiment	lin	RW
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW
GLB_USLR	US 10-Year Treasury Yield (global common)	lin	RW
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW
GLB_USBAA	US Baa Corporate Bond Yield, credit proxy (global common)	lin	RW
FXTWM	Nominal Trade-Weighted Exch Value of US$ vs AFE Currencies	log	RW
GLB_USEQ	US Equity Price Index (global common)	log	RW
PZTEXP	Spot Oil Price: West Texas Intermediate	log	RW
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW
HP	House Price	log	RW
LEV	Real nonfinancial business debt	log	RW
FREV_GDP	Federal Receipts (% of GDP)	lin	WN
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN

ID	Description	Trans.	Prior
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN
GLB_OIL	Spot Oil Price, WTI (global common)	log	RW
ROWDEM_US	US RoW demand (trade-weighted partner GDP growth)	lin	RW
ROWINF_US	US RoW inflation (trade-weighted partner inflation)	lin	RW
ROWFIN_US	US RoW financial conditions (trade-weighted partner rates)	lin	RW

The other coupled blocs follow the same estimation methodology with country-appropriate variable selections, replacing the US fiscal and detailed financial block with the three `ROW*` stars of Section 9.1. Across the deployed roster the partner blocs range from $n = 14$ (Costa Rica) to $n = 25$ (UK and Canada), with the euro-area aggregate at $n = 24$ and the core majors at $n \in \{18, \ldots, 25\}$ (e.g. CH, NO = 18, CN = 20, JP = 23); the US is the largest at $n = 41$. The China bloc uses a tighter Minnesota tightness to discipline its shorter, noisier sample.

9.8 Variable Transformations, Priors, and Financial Tags by Economy

Section 9.7 tabulates the US bloc in full; this subsection extends that detail to *every* deployed economy. For all 50 blocs (the 33 GVAR members of Section 9.1 and the 17 euro-area standalones of Section 9.2) each variable's *transformation* (`log` = log level, growth displayed; `lin` = level), its *financial* tag (which governs whether a series is held observed through the COVID window, Section 9.4), and its *prior* mean (RW = random walk in levels; WN = white noise / stationary) are read directly from the deployed `public/data/bvar/<CC>/spec.json` specifications, so the tables below mirror the current re-estimated state of the tool exactly.

Cross-economy coherence matrix. Table 4 collects the common concepts that recur across blocs. Because the transformation, financial tag, and prior for each concept are assigned *identically* in every economy that carries it, a single row characterises the concept tool-wide. Every variable is given a random-walk (RW) prior *except* the three fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`), which are bounded mean-reverting shares and therefore carry a white-noise / stationary (WN) prior; the effective-interest series `EFFI` remains RW.

The 33 GVAR blocs. Each carries a domestic block, the four global commons (`GLB_*`), and the three trade-weighted `ROW*_<CC>` stars. Tables follow in alphabetical order.

Table 4: Cross-economy coherence matrix for the common concepts. "Trans." is `log/lin`; "Fin." marks financial series (✓); "Prior" is RW (random walk) or WN (white noise / stationary). Each concept takes the same triple in every bloc that carries it.

Concept	**Trans.**	**Fin.**	**Prior**
Real GDP (level)	`log`	–	RW
Price index (CPI/HICP/deflator)	`log`	–	RW
Policy rate	`lin`	✓	RW
Short rate (3M/money-mkt)	`lin`	✓	RW
10Y government yield	`lin`	✓	RW
Corporate / Baa spread	`lin`	✓	RW
House prices	`log`	✓	RW
Equity price index	`log`	✓	RW
Oil price (WTI, global)	`log`	✓	RW
FX / NEER	`log`	✓	RW
Fiscal revenue ratio (FREV_GDP)	`lin`	–	WN
Fiscal primary-exp. ratio (FPEXP_GDP)	`lin`	–	WN
Stock-flow adjustment (SFA)	`lin`	–	WN
Effective interest (EFFI)	`lin`	✓	RW
RoW demand star (ROWDEM)	`lin`	✓	RW
RoW inflation star (ROWINF)	`lin`	✓	RW
RoW financial star (ROWFIN)	`lin`	✓	RW

Table 5: Argentina (`AR`, $n = 18$): variable transformations, financial tags, and priors.

ID	**Trans.**	**Fin.**	**Prior**
`GDP_AR`	`log`	–	RW
`CONS_AR`	`log`	–	RW
`INV_AR`	`log`	–	RW
`GOV_AR`	`log`	–	RW
`EXP_AR`	`log`	–	RW
`IMP_AR`	`log`	–	RW
`CPI_AR`	`log`	–	RW
`LR_AR`	`lin`	–	RW
`RATE_AR`	`lin`	✓	RW
`NEER_AR`	`log`	✓	RW
`POLRATE_AR`	`lin`	✓	RW
`GLB_OIL`	`log`	✓	RW
`GLB_USLR`	`lin`	✓	RW
`GLB_USBAA`	`lin`	✓	RW
`GLB_USEQ`	`log`	✓	RW
`ROWDEM_AR`	`lin`	✓	RW
`ROWINF_AR`	`lin`	✓	RW
`ROWFIN_AR`	`lin`	✓	RW

Table 6: Australia (AU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AU	log	–	RW
CONS_AU	log	–	RW
GOV_AU	log	–	RW
INV_AU	log	–	RW
EXP_AU	log	–	RW
IMP_AU	log	–	RW
CPI_AU	log	–	RW
LR_AU	lin	–	RW
NEER_AU	log	✓	RW
RATE_AU	lin	✓	RW
YIELD10_AU	lin	✓	RW
HP_AU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_AU	lin	✓	RW
ROWINF_AU	lin	✓	RW
ROWFIN_AU	lin	✓	RW

Table 7: Brazil (BR, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_BR	log	–	RW
CONS_BR	log	–	RW
INV_BR	log	–	RW
GOV_BR	log	–	RW
EXP_BR	log	–	RW
IMP_BR	log	–	RW
CPI_BR	log	–	RW
LR_BR	lin	–	RW
NEER_BR	log	✓	RW
POLRATE_BR	lin	✓	RW
HP_BR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_BR	lin	✓	RW
ROWINF_BR	lin	✓	RW
ROWFIN_BR	lin	✓	RW

Table 8: Canada (CA, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CA	log	–	RW
CONS_CA	log	–	RW
INV_CA	log	–	RW
GOV_CA	log	–	RW
EXP_CA	log	–	RW
IMP_CA	log	–	RW
IP_CA	log	–	RW
LR_CA	lin	–	RW
CPI_CA	log	–	RW
RATE_CA	lin	✓	RW
POLRATE_CA	lin	✓	RW
YIELD10_CA	lin	✓	RW
HP_CA	log	✓	RW
NEER_CA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CA	lin	✓	RW
ROWINF_CA	lin	✓	RW
ROWFIN_CA	lin	✓	RW

Table 9: Switzerland (CH, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CH	log	–	RW
CONS_CH	log	–	RW
INV_CH	log	–	RW
GOV_CH	log	–	RW
EXP_CH	log	–	RW
IMP_CH	log	–	RW
HICP_CH	log	–	RW
LR_CH	lin	–	RW
NEER_CH	log	✓	RW
POLRATE_CH	lin	✓	RW
HP_CH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW

Switzerland (CH), continued

ID	Trans.	Fin.	Prior
GLB_USEQ	log	✓	RW
ROWDEM_CH	lin	✓	RW
ROWINF_CH	lin	✓	RW
ROWFIN_CH	lin	✓	RW

Table 10: Chile (CL, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CL	log	–	RW
CONS_CL	log	–	RW
INV_CL	log	–	RW
GOV_CL	log	–	RW
EXP_CL	log	–	RW
IMP_CL	log	–	RW
CPI_CL	log	–	RW
LR_CL	lin	–	RW
RATE_CL	lin	✓	RW
POLRATE_CL	lin	✓	RW
YIELD10_CL	lin	✓	RW
HP_CL	log	✓	RW
NEER_CL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CL	lin	✓	RW
ROWINF_CL	lin	✓	RW
ROWFIN_CL	lin	✓	RW

Table 11: China (CN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CN	log	–	RW
CPI_CN	log	–	RW
RATE_CN	lin	✓	RW
EXP_CN	log	–	RW
IMP_CN	log	–	RW
REER_CN	log	✓	RW
EQ_CN	log	✓	RW
POLRATE_CN	lin	✓	RW
HP_CN	log	✓	RW
GLB_OIL	log	✓	RW

China (CN), continued

ID	Trans.	Fin.	Prior
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CN	lin	✓	RW
ROWINF_CN	lin	✓	RW
ROWFIN_CN	lin	✓	RW

Table 12: Colombia (CO, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CO	log	–	RW
CONS_CO	log	–	RW
INV_CO	log	–	RW
GOV_CO	log	–	RW
EXP_CO	log	–	RW
IMP_CO	log	–	RW
CPI_CO	log	–	RW
LR_CO	lin	–	RW
RATE_CO	lin	✓	RW
POLRATE_CO	lin	✓	RW
YIELD10_CO	lin	✓	RW
HP_CO	log	✓	RW
NEER_CO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CO	lin	✓	RW
ROWINF_CO	lin	✓	RW
ROWFIN_CO	lin	✓	RW

Table 13: Costa Rica (CR, $n = 14$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CR	log	–	RW
EXP_CR	log	–	RW
IMP_CR	log	–	RW
CPI_CR	log	–	RW

Costa Rica (CR), continued

ID	Trans.	Fin.	Prior
LR_CR	lin	–	RW
RATE_CR	lin	✓	RW
NEER_CR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CR	lin	✓	RW
ROWINF_CR	lin	✓	RW
ROWFIN_CR	lin	✓	RW

Table 14: Czechia (CZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CZ	log	–	RW
CONS_CZ	log	–	RW
INV_CZ	log	–	RW
GOV_CZ	log	–	RW
EXP_CZ	log	–	RW
IMP_CZ	log	–	RW
HICP_CZ	log	–	RW
LR_CZ	lin	–	RW
YIELD10_CZ	lin	✓	RW
NEER_CZ	log	✓	RW
POLRATE_CZ	lin	✓	RW
HP_CZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CZ	lin	✓	RW
ROWINF_CZ	lin	✓	RW
ROWFIN_CZ	lin	✓	RW

Table 15: Denmark (DK, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DK	log	–	RW
CONS_DK	log	–	RW
INV_DK	log	–	RW
GOV_DK	log	–	RW
EXP_DK	log	–	RW

Denmark (DK), continued

ID	Trans.	Fin.	Prior
IMP_DK	log	–	RW
HICP_DK	log	–	RW
LR_DK	lin	–	RW
RATE_DK	lin	✓	RW
YIELD10_DK	lin	✓	RW
NEER_DK	log	✓	RW
POLRATE_DK	lin	✓	RW
HP_DK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_DK	lin	✓	RW
ROWINF_DK	lin	✓	RW
ROWFIN_DK	lin	✓	RW

Table 16: Euro Area (EA, $n = 24$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EA	log	–	RW
CONS_EA	log	–	RW
INV_EA	log	–	RW
GOV_EA	log	–	RW
EXP_EA	log	–	RW
IMP_EA	log	–	RW
HICP_EA	log	–	RW
RATE_EA	lin	✓	RW
YIELD10_EA	lin	✓	RW
NEER_EA	log	✓	RW
POLRATE_EA	lin	✓	RW
HPI_EA	log	✓	RW
EACORP	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_EA	lin	✓	RW
ROWINF_EA	lin	✓	RW
ROWFIN_EA	lin	✓	RW

Table 17: Hungary (HU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_HU	log	–	RW
CONS_HU	log	–	RW
INV_HU	log	–	RW
GOV_HU	log	–	RW
EXP_HU	log	–	RW
IMP_HU	log	–	RW
HICP_HU	log	–	RW
LR_HU	lin	–	RW
YIELD10_HU	lin	✓	RW
NEER_HU	log	✓	RW
POLRATE_HU	lin	✓	RW
HP_HU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_HU	lin	✓	RW
ROWINF_HU	lin	✓	RW
ROWFIN_HU	lin	✓	RW

Table 18: Indonesia (ID, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ID	log	–	RW
CONS_ID	log	–	RW
INV_ID	log	–	RW
GOV_ID	log	–	RW
EXP_ID	log	–	RW
IMP_ID	log	–	RW
CPI_ID	log	–	RW
RATE_ID	lin	✓	RW
NEER_ID	log	✓	RW
POLRATE_ID	lin	✓	RW
HP_ID	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ID	lin	✓	RW
ROWINF_ID	lin	✓	RW
ROWFIN_ID	lin	✓	RW

Table 19: Israel (IL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IL	log	–	RW
CONS_IL	log	–	RW
INV_IL	log	–	RW
GOV_IL	log	–	RW
EXP_IL	log	–	RW
IMP_IL	log	–	RW
CPI_IL	log	–	RW
LR_IL	lin	–	RW
RATE_IL	lin	✓	RW
POLRATE_IL	lin	✓	RW
NEER_IL	log	✓	RW
HP_IL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IL	lin	✓	RW
ROWINF_IL	lin	✓	RW
ROWFIN_IL	lin	✓	RW

Table 20: India (IN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IN	log	–	RW
CONS_IN	log	–	RW
INV_IN	log	–	RW
GOV_IN	log	–	RW
EXP_IN	log	–	RW
IMP_IN	log	–	RW
CPI_IN	log	–	RW
RATE_IN	lin	✓	RW
RATE3M_IN	lin	✓	RW
YIELD10_IN	lin	✓	RW
NEER_IN	log	✓	RW
POLRATE_IN	lin	✓	RW
HP_IN	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IN	lin	✓	RW
ROWINF_IN	lin	✓	RW
ROWFIN_IN	lin	✓	RW

Table 21: Iceland (IS, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IS	log	–	RW
CONS_IS	log	–	RW
INV_IS	log	–	RW
GOV_IS	log	–	RW
EXP_IS	log	–	RW
IMP_IS	log	–	RW
CPI_IS	log	–	RW
LR_IS	lin	–	RW
RATE_IS	lin	✓	RW
NEER_IS	log	✓	RW
POLRATE_IS	lin	✓	RW
HP_IS	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IS	lin	✓	RW
ROWINF_IS	lin	✓	RW
ROWFIN_IS	lin	✓	RW

Table 22: Japan (JP, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_JP	log	–	RW
CONS_JP	log	–	RW
INV_JP	log	–	RW
GOV_JP	log	–	RW
EXP_JP	log	–	RW
IMP_JP	log	–	RW
LR_JP	lin	–	RW
CPI_JP	log	–	RW
RATE_JP	lin	✓	RW
POLRATE_JP	lin	✓	RW
YIELD10_JP	lin	✓	RW
HP_JP	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN

Japan (JP), continued

ID	Trans.	Fin.	Prior
ROWDEM_JP	lin	✓	RW
ROWINF_JP	lin	✓	RW
ROWFIN_JP	lin	✓	RW

Table 23: Korea (KR, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_KR	log	–	RW
CONS_KR	log	–	RW
INV_KR	log	–	RW
GOV_KR	log	–	RW
EXP_KR	log	–	RW
IMP_KR	log	–	RW
CPI_KR	log	–	RW
LR_KR	lin	–	RW
RATE_KR	lin	✓	RW
POLRATE_KR	lin	✓	RW
YIELD10_KR	lin	✓	RW
HP_KR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_KR	lin	✓	RW
ROWINF_KR	lin	✓	RW
ROWFIN_KR	lin	✓	RW

Table 24: Mexico (MX, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MX	log	–	RW
CONS_MX	log	–	RW
INV_MX	log	–	RW
GOV_MX	log	–	RW
EXP_MX	log	–	RW
IMP_MX	log	–	RW
CPI_MX	log	–	RW
LR_MX	lin	–	RW
RATE_MX	lin	✓	RW
POLRATE_MX	lin	✓	RW
YIELD10_MX	lin	✓	RW
HP_MX	log	✓	RW

Mexico (MX), continued

ID	Trans.	Fin.	Prior
NEER_MX	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MX	lin	✓	RW
ROWINF_MX	lin	✓	RW
ROWFIN_MX	lin	✓	RW

Table 25: Malaysia (MY, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MY	log	–	RW
CONS_MY	log	–	RW
INV_MY	log	–	RW
GOV_MY	log	–	RW
EXP_MY	log	–	RW
IMP_MY	log	–	RW
CPI_MY	log	–	RW
LR_MY	lin	–	RW
RATE_MY	lin	✓	RW
NEER_MY	log	✓	RW
POLRATE_MY	lin	✓	RW
HP_MY	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MY	lin	✓	RW
ROWINF_MY	lin	✓	RW
ROWFIN_MY	lin	✓	RW

Table 26: Norway (NO, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NO	log	–	RW
CONS_NO	log	–	RW
INV_NO	log	–	RW
GOV_NO	log	–	RW
EXP_NO	log	–	RW
IMP_NO	log	–	RW
HICP_NO	log	–	RW

Norway (NO), continued

ID	Trans.	Fin.	Prior
LR_NO	lin	–	RW
NEER_NO	log	✓	RW
POLRATE_NO	lin	✓	RW
HP_NO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NO	lin	✓	RW
ROWINF_NO	lin	✓	RW
ROWFIN_NO	lin	✓	RW

Table 27: New Zealand (NZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NZ	log	–	RW
CONS_NZ	log	–	RW
INV_NZ	log	–	RW
GOV_NZ	log	–	RW
EXP_NZ	log	–	RW
IMP_NZ	log	–	RW
CPI_NZ	log	–	RW
LR_NZ	lin	–	RW
RATE_NZ	lin	✓	RW
YIELD10_NZ	lin	✓	RW
HP_NZ	log	✓	RW
NEER_NZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NZ	lin	✓	RW
ROWINF_NZ	lin	✓	RW
ROWFIN_NZ	lin	✓	RW

Table 28: Poland (PL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PL	log	–	RW
CONS_PL	log	–	RW
INV_PL	log	–	RW
GOV_PL	log	–	RW

Poland (PL), continued

ID	Trans.	Fin.	Prior
EXP_PL	log	–	RW
IMP_PL	log	–	RW
HICP_PL	log	–	RW
LR_PL	lin	–	RW
YIELD10_PL	lin	✓	RW
NEER_PL	log	✓	RW
POLRATE_PL	lin	✓	RW
HP_PL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_PL	lin	✓	RW
ROWINF_PL	lin	✓	RW
ROWFIN_PL	lin	✓	RW

Table 29: Russia (RU, $n = 15$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_RU	log	–	RW
CPI_RU	log	–	RW
EXP_RU	log	–	RW
LR_RU	lin	–	RW
RATE_RU	lin	✓	RW
NEER_RU	log	✓	RW
POLRATE_RU	lin	✓	RW
HP_RU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_RU	lin	✓	RW
ROWINF_RU	lin	✓	RW
ROWFIN_RU	lin	✓	RW

Table 30: Saudi Arabia (SA, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SA	log	–	RW
CONS_SA	log	–	RW
INV_SA	log	–	RW
GOV_SA	log	–	RW

Saudi Arabia (SA), continued

ID	Trans.	Fin.	Prior
EXP_SA	log	–	RW
IMP_SA	log	–	RW
CPI_SA	log	–	RW
LR_SA	lin	–	RW
RATE_SA	lin	✓	RW
NEER_SA	log	✓	RW
POLRATE_SA	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SA	lin	✓	RW
ROWINF_SA	lin	✓	RW
ROWFIN_SA	lin	✓	RW

Table 31: Sweden (SE, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SE	log	–	RW
CONS_SE	log	–	RW
INV_SE	log	–	RW
GOV_SE	log	–	RW
EXP_SE	log	–	RW
IMP_SE	log	–	RW
HICP_SE	log	–	RW
LR_SE	lin	–	RW
RATE_SE	lin	✓	RW
YIELD10_SE	lin	✓	RW
NEER_SE	log	✓	RW
POLRATE_SE	lin	✓	RW
HP_SE	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SE	lin	✓	RW
ROWINF_SE	lin	✓	RW
ROWFIN_SE	lin	✓	RW

Table 32: Thailand (TH, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TH	log	–	RW
CONS_TH	log	–	RW
INV_TH	log	–	RW
GOV_TH	log	–	RW
EXP_TH	log	–	RW
IMP_TH	log	–	RW
CPI_TH	log	–	RW
LR_TH	lin	–	RW
RATE_TH	lin	✓	RW
YIELD10_TH	lin	✓	RW
NEER_TH	log	✓	RW
POLRATE_TH	lin	✓	RW
HP_TH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TH	lin	✓	RW
ROWINF_TH	lin	✓	RW
ROWFIN_TH	lin	✓	RW

Table 33: Türkiye (TR, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TR	log	–	RW
CONS_TR	log	–	RW
INV_TR	log	–	RW
GOV_TR	log	–	RW
EXP_TR	log	–	RW
IMP_TR	log	–	RW
CPI_TR	log	–	RW
LR_TR	lin	–	RW
RATE_TR	lin	✓	RW
POLRATE_TR	lin	✓	RW
HP_TR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TR	lin	✓	RW
ROWINF_TR	lin	✓	RW
ROWFIN_TR	lin	✓	RW

Table 34: Taiwan (TW, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TW	log	–	RW
CONS_TW	log	–	RW
GOV_TW	log	–	RW
INV_TW	log	–	RW
EXP_TW	log	–	RW
IMP_TW	log	–	RW
CPI_TW	log	–	RW
LR_TW	lin	–	RW
RATE_TW	lin	✓	RW
MMRATE_TW	lin	✓	RW
YIELD10_TW	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TW	lin	✓	RW
ROWINF_TW	lin	✓	RW
ROWFIN_TW	lin	✓	RW

Table 35: United Kingdom (UK, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_UK	log	–	RW
CONS_UK	log	–	RW
INV_UK	log	–	RW
GOV_UK	log	–	RW
EXP_UK	log	–	RW
IMP_UK	log	–	RW
IP_UK	log	–	RW
LR_UK	lin	–	RW
CPI_UK	log	–	RW
RATE_UK	lin	✓	RW
POLRATE_UK	lin	✓	RW
YIELD10_UK	lin	✓	RW
HP_UK	log	✓	RW
NEER_UK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United Kingdom (UK), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
ROWDEM_UK	lin	✓	RW
ROWINF_UK	lin	✓	RW
ROWFIN_UK	lin	✓	RW

Table 36: United States (US, $n = 41$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDPH	log	–	RW
CH	log	–	RW
FRH	log	–	RW
FNH	log	–	RW
XH	log	–	RW
MH	log	–	RW
GH	log	–	RW
JGDP	log	–	RW
JC	log	–	RW
JCXFE	log	–	RW
UI	log	–	RW
UIXFDG	log	–	RW
LXBC	log	–	RW
LANAGRA	log	–	RW
LR	lin	–	RW
IP	log	–	RW
CUMFG	lin	–	RW
HST	log	–	RW
YPDH	log	–	RW
CSENT	lin	–	RW
POLRATE	lin	✓	RW
FCM2	lin	✓	RW
FCM5	lin	✓	RW
GLB_USLR	lin	✓	RW
FAAA	lin	✓	RW
GLB_USBAA	lin	✓	RW
FXTWM	log	✓	RW
GLB_USEQ	log	✓	RW
PZTEXP	log	✓	RW
GSNE	log	✓	RW
SP500VOL	lin	✓	RW
HP	log	✓	RW
LEV	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United States (US), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
GLB_OIL	log	✓	RW
ROWDEM_US	lin	✓	RW
ROWINF_US	lin	✓	RW
ROWFIN_US	lin	✓	RW

Table 37: South Africa (ZA, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ZA	log	–	RW
CONS_ZA	log	–	RW
INV_ZA	log	–	RW
GOV_ZA	log	–	RW
EXP_ZA	log	–	RW
IMP_ZA	log	–	RW
CPI_ZA	log	–	RW
LR_ZA	lin	–	RW
RATE_ZA	lin	✓	RW
YIELD10_ZA	lin	✓	RW
NEER_ZA	log	✓	RW
POLRATE_ZA	lin	✓	RW
HP_ZA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ZA	lin	✓	RW
ROWINF_ZA	lin	✓	RW
ROWFIN_ZA	lin	✓	RW

The 17 euro-area standalones. These omit the ROW* stars and instead carry US-block foreign variables (*_US) plus the shared euro-area corporate spread (EACORP); they are estimated in separation (Section 9.2).

Table 38: Austria (AT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AT	log	–	RW
CONS_AT	log	–	RW
INV_AT	log	–	RW
GOV_AT	log	–	RW
EXP_AT	log	–	RW

Austria (AT), continued

ID	Trans.	Fin.	Prior
IMP_AT	log	–	RW
HICP_AT	log	–	RW
LR_AT	lin	–	RW
POLRATE_AT	lin	✓	RW
HPI_AT	log	✓	RW
EACORP	lin	✓	RW
RATE_AT	lin	✓	RW
YIELD10_AT	lin	✓	RW
NEER_AT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 39: Belgium (BE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_BE	log	–	RW
CONS_BE	log	–	RW
INV_BE	log	–	RW
GOV_BE	log	–	RW
EXP_BE	log	–	RW
IMP_BE	log	–	RW
HICP_BE	log	–	RW
LR_BE	lin	–	RW
POLRATE_BE	lin	✓	RW
HPI_BE	log	✓	RW
EACORP	lin	✓	RW
RATE_BE	lin	✓	RW
YIELD10_BE	lin	✓	RW
NEER_BE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Belgium (BE), continued

ID	Trans.	Fin.	Prior
WTI_US	log	✓	RW

Table 40: Germany (DE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DE	log	–	RW
CONS_DE	log	–	RW
INV_DE	log	–	RW
GOV_DE	log	–	RW
EXP_DE	log	–	RW
IMP_DE	log	–	RW
HICP_DE	log	–	RW
LR_DE	lin	–	RW
POLRATE_DE	lin	✓	RW
HPI_DE	log	✓	RW
EACORP	lin	✓	RW
RATE_DE	lin	✓	RW
YIELD10_DE	lin	✓	RW
NEER_DE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 41: Estonia (EE, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EE	log	–	RW
CONS_EE	log	–	RW
INV_EE	log	–	RW
GOV_EE	log	–	RW
EXP_EE	log	–	RW
IMP_EE	log	–	RW
HICP_EE	log	–	RW
LR_EE	lin	–	RW
POLRATE_EE	lin	✓	RW
HPI_EE	log	✓	RW

Estonia (EE), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_EE	lin	✓	RW
NEER_EE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 42: Spain (ES, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ES	log	–	RW
CONS_ES	log	–	RW
INV_ES	log	–	RW
GOV_ES	log	–	RW
EXP_ES	log	–	RW
IMP_ES	log	–	RW
HICP_ES	log	–	RW
LR_ES	lin	–	RW
POLRATE_ES	lin	✓	RW
HPI_ES	log	✓	RW
EACORP	lin	✓	RW
RATE_ES	lin	✓	RW
YIELD10_ES	lin	✓	RW
NEER_ES	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 43: Finland (FI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FI	log	–	RW
CONS_FI	log	–	RW
INV_FI	log	–	RW
GOV_FI	log	–	RW
EXP_FI	log	–	RW
IMP_FI	log	–	RW
HICP_FI	log	–	RW
LR_FI	lin	–	RW
POLRATE_FI	lin	✓	RW
HPI_FI	log	✓	RW
EACORP	lin	✓	RW
RATE_FI	lin	✓	RW
YIELD10_FI	lin	✓	RW
NEER_FI	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 44: France (FR, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FR	log	–	RW
CONS_FR	log	–	RW
INV_FR	log	–	RW
GOV_FR	log	–	RW
EXP_FR	log	–	RW
IMP_FR	log	–	RW
HICP_FR	log	–	RW
LR_FR	lin	–	RW
POLRATE_FR	lin	✓	RW
HPI_FR	log	✓	RW
EACORP	lin	✓	RW
RATE_FR	lin	✓	RW
YIELD10_FR	lin	✓	RW
NEER_FR	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW

France (FR), continued

ID	Trans.	Fin.	Prior
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 45: Greece (GR, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_GR	log	–	RW
CONS_GR	log	–	RW
INV_GR	log	–	RW
GOV_GR	log	–	RW
EXP_GR	log	–	RW
IMP_GR	log	–	RW
HICP_GR	log	–	RW
LR_GR	lin	–	RW
POLRATE_GR	lin	✓	RW
EACORP	lin	✓	RW
RATE_GR	lin	✓	RW
YIELD10_GR	lin	✓	RW
NEER_GR	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 46: Ireland (IE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IE	log	–	RW
CONS_IE	log	–	RW
INV_IE	log	–	RW
GOV_IE	log	–	RW
EXP_IE	log	–	RW

Ireland (IE), continued

ID	Trans.	Fin.	Prior
IMP_IE	log	–	RW
HICP_IE	log	–	RW
LR_IE	lin	–	RW
POLRATE_IE	lin	✓	RW
HPI_IE	log	✓	RW
EACORP	lin	✓	RW
RATE_IE	lin	✓	RW
YIELD10_IE	lin	✓	RW
NEER_IE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 47: Italy (IT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IT	log	–	RW
CONS_IT	log	–	RW
INV_IT	log	–	RW
GOV_IT	log	–	RW
EXP_IT	log	–	RW
IMP_IT	log	–	RW
HICP_IT	log	–	RW
LR_IT	lin	–	RW
POLRATE_IT	lin	✓	RW
HPI_IT	log	✓	RW
EACORP	lin	✓	RW
RATE_IT	lin	✓	RW
YIELD10_IT	lin	✓	RW
NEER_IT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Italy (IT), continued

ID	Trans.	Fin.	Prior
WTI_US	log	✓	RW

Table 48: Lithuania (LT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LT	log	–	RW
CONS_LT	log	–	RW
INV_LT	log	–	RW
GOV_LT	log	–	RW
EXP_LT	log	–	RW
IMP_LT	log	–	RW
HICP_LT	log	–	RW
LR_LT	lin	–	RW
POLRATE_LT	lin	✓	RW
HPI_LT	log	✓	RW
EACORP	lin	✓	RW
RATE_LT	lin	✓	RW
YIELD10_LT	lin	✓	RW
NEER_LT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 49: Luxembourg (LU, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LU	log	–	RW
CONS_LU	log	–	RW
INV_LU	log	–	RW
GOV_LU	log	–	RW
EXP_LU	log	–	RW
IMP_LU	log	–	RW
HICP_LU	log	–	RW
LR_LU	lin	–	RW
POLRATE_LU	lin	✓	RW
HPI_LU	log	✓	RW

Luxembourg (LU), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_LU	lin	✓	RW
YIELD10_LU	lin	✓	RW
NEER_LU	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 50: Latvia (LV, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LV	log	–	RW
CONS_LV	log	–	RW
INV_LV	log	–	RW
GOV_LV	log	–	RW
EXP_LV	log	–	RW
IMP_LV	log	–	RW
HICP_LV	log	–	RW
LR_LV	lin	–	RW
RATE_LV	lin	✓	RW
YIELD10_LV	lin	✓	RW
NEER_LV	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 51: Netherlands (NL, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NL	log	–	RW

Netherlands (NL), continued

ID	Trans.	Fin.	Prior
CONS_NL	log	–	RW
INV_NL	log	–	RW
GOV_NL	log	–	RW
EXP_NL	log	–	RW
IMP_NL	log	–	RW
HICP_NL	log	–	RW
LR_NL	lin	–	RW
POLRATE_NL	lin	✓	RW
HPI_NL	log	✓	RW
EACORP	lin	✓	RW
RATE_NL	lin	✓	RW
YIELD10_NL	lin	✓	RW
NEER_NL	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 52: Portugal (PT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PT	log	–	RW
CONS_PT	log	–	RW
INV_PT	log	–	RW
GOV_PT	log	–	RW
EXP_PT	log	–	RW
IMP_PT	log	–	RW
HICP_PT	log	–	RW
LR_PT	lin	–	RW
POLRATE_PT	lin	✓	RW
HPI_PT	log	✓	RW
EACORP	lin	✓	RW
RATE_PT	lin	✓	RW
YIELD10_PT	lin	✓	RW
NEER_PT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW

Portugal (PT), continued

ID	Trans.	Fin.	Prior
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 53: Slovenia (SI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SI	log	–	RW
CONS_SI	log	–	RW
INV_SI	log	–	RW
GOV_SI	log	–	RW
EXP_SI	log	–	RW
IMP_SI	log	–	RW
HICP_SI	log	–	RW
LR_SI	lin	–	RW
POLRATE_SI	lin	✓	RW
HPI_SI	log	✓	RW
EACORP	lin	✓	RW
RATE_SI	lin	✓	RW
YIELD10_SI	lin	✓	RW
NEER_SI	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 54: Slovakia (SK, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SK	log	–	RW
CONS_SK	log	–	RW
INV_SK	log	–	RW
GOV_SK	log	–	RW
EXP_SK	log	–	RW
IMP_SK	log	–	RW

Slovakia (SK), continued

ID	Trans.	Fin.	Prior
HICP_SK	log	–	RW
LR_SK	lin	–	RW
POLRATE_SK	lin	✓	RW
HPI_SK	log	✓	RW
EACORP	lin	✓	RW
RATE_SK	lin	✓	RW
YIELD10_SK	lin	✓	RW
NEER_SK	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Coherence review. A cross-economy audit of all 1048 variable rows confirms that every shared concept is assigned the same transformation, financial tag, and prior in every bloc that carries it: real activity, price indices, house prices, equities, oil, and FX enter in `log`; interest rates, yields, spreads, unemployment, and the fiscal ratios enter `lin`; all financial series share the financial tag; and all priors are RW apart from the three stationary fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`, prior WN). Canada's industrial-production series `IP_CA` has been rebuilt as a `log`-level index, consistent with the activity convention used for the United States and the United Kingdom. The FRED level index (`CANPROINDMISMEI`, 2015 = 100) was discontinued at 2024-02, so the series is the quarterly average of that index through 2024-Q1 and is then growth-chained over 2024-Q2–2025-Q4 (eight quarters) using the FRED q-on-q growth series `PRINTO01CAQ657S` ($\text{level}_t = \text{level}_{t-1}(1 + g_t)$, the same chaining used to extend the other Canadian series). With this reconstruction the audit finds no remaining transformation-coherence exceptions.

10 User Interface Guide

10.1 Top Bar

The tool is deployed as a static site served at the `/bvar-scenario` path of the academic website, which is hosted on **GitHub Pages** (the `matyasfarkas/starter-hugo-academic` Hugo Academic site). The top bar provides global controls:

- **Country selector:** Switch between any of the roughly fifty estimated economies. The drop-down is grouped into two sections by actual GVAR status: a *GVAR* group listing the 33 coupled blocs, and an *Other countries* group listing the standalone single-country models (Section 9). Membership is conveyed purely by the section a country sits in—the names are shown plainly, with no per-option suffix or bold—and the split is driven by the live trade-weights roster rather than a static geographic grouping. Switching reloads all BVAR data and resets conditioning.

- **User Manual:** Download this document as a PDF.
- **Uncertainty mode:** "None" (deterministic), "Quick" ($D = 50$ draws), or "Full" ($D = 200$ draws).
- **Compute button:** Triggers the simulation smoother (Section 4.6). For the long-run anchor (the entropic tilt of Section 8.2), the Compute button is enabled whenever an anchor is active and auto-selects "Quick" mode if no uncertainty mode is chosen.

10.2 Chart Panel

The main panel displays a grid of time-series charts. The first row shows the primary macroeconomic triad—inflation, policy rate, GDP growth—plus government spending. Subsequent rows show core inflation, unemployment, consumption, investment, and any country-specific variables.

Each chart includes:

- A **dotted baseline** forecast with 68% credible bands (shaded).
- A **solid scenario line** showing the conditional forecast.
- A **dashed counterfactual line** (when active) showing the policy counterfactual.
- A **"Condition" button** to toggle conditioning on the variable.
- **Drag interaction:** Click and drag data points on conditioned charts to set target paths. Right-click to toggle individual quarters on or off.

GVAR solve badge and the "Solve full GVAR" button. On a GVAR-linked bloc a small badge above the charts reports the cross-country solve that produced the displayed scenario, together with its sweep count, spectral radius ρ, and time. Because the US is now endogenous (Section 6.5), dragging a pin shows a fast first-order *direct preview* (an amber "GVAR DIRECT preview" badge); when a preview is on screen a blue **Solve full GVAR** button is offered to compute the complete cross-country fixed point—all feedback loops, including the Rest-of-World feedback onto the US (a blue "GVAR full solve" badge). While a solve is running a "computing..." pill is shown so a structure-change re-probe does not look like a freeze. On standalone (non-GVAR) blocs no badge or button appears; the scenario is the ordinary single-country conditional forecast.

10.3 Control Panel (Sidebar)

10.3.1 Scenario Section

- **Scenario name:** Editable text field.
- **Reset:** Clear all conditioning to return to baseline.
- **Save/Load:** Export and import scenarios as Excel files. The Excel format includes sheets for settings, conditioning paths, active quarters, rate adjustments, and a variable map for reproducibility.
- **Example:** Download a pre-built inflation-surge scenario.
- **Conditioning sliders:** Per-variable, per-quarter deviation sliders (± 3 pp) for each conditioned variable.

10.3.2 Policy Counterfactual Section

- **Counterfactual toggle:** Enter/exit counterfactual mode.
- **DSGE model browser:** Select from 169 models, searchable by name or reference.
- **Policy rule selector:** Choose from 9 standardized monetary policy rules (Table in Section 5.1).
- **Rule-based / Optimal toggle:** Switch between specifying the rate path via a Taylor-type rule or via the Barnichon and Mesters (2023) optimal policy (Section 5.4).
- **Dual-mandate slider:** In optimal mode, adjust the relative weight on price stability versus full employment.

10.4 Export

The scenario bar at the bottom provides four export formats:

- **CSV:** Tabular data with columns for each variable's Q16 (16th percentile), baseline (median), Q84 (84th percentile), active scenario, deviation from baseline, and counterfactual (if active). Rows span historical quarters through the full forecast horizon.
- **Excel:** A workbook with per-variable detail sheets (Q16, baseline, Q84, scenario, deviation, counterfactual) and a summary sheet showing all variables side by side.
- **PNG:** A ZIP archive of high-resolution chart images (1200×500 px) capturing the full visual state including all rendered scenarios, uncertainty bands, and counterfactuals.
- **PDF:** A multi-page PDF with one chart per page (landscape orientation), including metadata footers with country, timestamp, and chart title.

All tabular exports (CSV, Excel) respect the active growth-rate toggle: if year-over-year is selected, the exported values reflect the y/y transformation. Visual exports (PNG, PDF) capture exactly what is displayed on screen, including all visible scenarios and their uncertainty bands.

10.4.1 Multi-Scenario Support

The tool supports up to four simultaneous scenarios. Each scenario's conditional forecast and uncertainty bands are cached when computed, so that switching between scenarios preserves their results. Tabular exports include data for the active scenario; visual exports capture all visible scenario lines and bands.

11 Technical Notes

11.1 Client-Side Computation

All matrix algebra, Kalman filtering, and simulation smoothing are performed in the browser using TypeScript with `Float64Array` storage. The linear algebra kernel implements:

- LU decomposition with partial pivoting for F_t^{-1} (singularity threshold: 10^{-15}).
- Cholesky decomposition for shock draws ($\Sigma = LL'$; diagonal check threshold: 10^{-14}).
- Row-major storage for cache-friendly matrix operations.

11.2 Performance Optimizations

- The unconditional smoother result is cached by a fingerprint of the model dimensions and a sample of coefficient values, avoiding recomputation when only the conditioning paths change.
- Conditioning input is debounced at 30 ms to avoid recomputation during interactive drag operations.
- The 500 thinned posterior draws ($\approx$34 MB for the US model) are loaded lazily—only when the user clicks "Compute."
- Plotly.js is loaded via dynamic `import()` to avoid blocking initial page render.

11.3 Interactive GVAR Architecture: keeping the heavy solve off the drag path

The deployed 50-bloc, US-endogenous GVAR poses a hard interactivity problem: a single coupled mean solve costs $\approx$ 4.6 s warm and $\approx$ 13 s cold (every US-pinned structure misses the shipped $(I - M)^{-1}$ coupling cache and must re-probe the Durbin–Koopman smoother per coupled bloc). Firing that solve on every pointer-move during a chart drag would make the interface unusable. The deployed design therefore removes the solve from the live drag path entirely, and reconciles the exact answer afterwards.

No solve during drag; exactly one cheap solve on release. A module-level signal bus (`gvarDragState.ts`) carries the reactive drag flag between the chart drag handler and the GVAR solve hook (`useGvarForecast.ts`), kept *out* of the React/Zustand store so that a pointer-down does not re-render the grid. While a drag is in progress *no* GVAR solve fires—neither a value-only nor a structure-change solve, since both are multi-second on the live panel. On `pointer-up` a single solve is dispatched after a short (100 ms) release macrotask that also coalesces back-to-back click-drag-click gestures into one dispatch. That release solve uses the cheap `'direct'` channel (one-hop local-elasticity preview, no 95 MB coupling-artifact structured-clone into the worker); the full cross-country fixed point with uncertainty bands (the `'full'` channel) runs *only* on the explicit "Compute" / "Solve full GVAR" action. Non-drag edits (numeric entry, Reset, scenario switch) still dispatch live: full solves run immediately and direct previews debounce (≈ 120 ms).

Optimistic conditioned-point dot. So that the user sees *immediate* feedback while the line and bands reconcile, the pinned marker (the conditioned-point "dot") is sourced from the *conditioning path* the click writes—not from the lagging solve output. The mapping lives in `conditionedDotY.ts`: `conditioningDeviationFromDisplayY` converts a clicked display y-value into the stored-space deviation, and its exact inverse `conditionedDotDisplayY` reconstructs the displayed dot, so the dot round-trips to the cursor to machine precision (`conditionedDotDisplayY`∘`conditioningDeviationFromDisplayY` id). The same inverse pair backs the right-click "plant the bullet at the cursor" gesture. The smoothed forecast line and the uncertainty bands then catch up when the solve returns.

Per-chart render memoization. The render cost of a conditioning edit was historically dominated by the *render*, not the solve: a monolithic chart component repainted all ≈ 35 panels on any edit. The deployed grid (`ChartGrid.tsx`) wraps each panel in `React.memo` and reconciles the freshly-built chart objects against the prior render with a tight structural equality (`reconcileCharts`/`chartDeepEqual`): a panel whose trace data did not move keeps its *prior object reference*, so `React.memo` skips it. A single conditioning edit therefore rebuilds and repaints *exactly one* of the ≈ 35 panels rather than all of them (asserted in `chartgrid-render-memo.test.ts`), with a stale-guard that still repaints every affected panel when the solve lands. The whole interactive contract—no solve on drag, one cheap solve on release, an optimistic dot, and single-panel repaints—is covered by the `gvar-drag-scheduling`, `conditioned-dot-optimistic`, `right-click-plant-at-cursor`, and `chartgrid-render-memo` test suites.

11.4 Numerical Conventions

- Measurement noise: $H = 10^{-12}\, I_n$, ensuring the Kalman filter treats conditioned values as near-exact observations while maintaining numerical stability.
- Growth-to-level conversion: $\Delta\ell = \Delta g/4$ for log-transformed variables, reflecting the annualization factor in (2).

- Tikhonov regularization for Barnichon–Mesters: $\alpha = 10^{-6}$.
- All matrix operations use IEEE 754 double precision (64-bit floating point).

11.5 Identifiability and Weak Exogeneity

The platform's separable estimation–solution architecture rests on a chain of identifying assumptions; we state them explicitly and are candid about where each binds.

Weak exogeneity of the star variables. Each bloc is estimated *conditionally* on its three trade-weighted star variables (Section 6.2), treating the foreign environment as predetermined. This is the weak-exogeneity device of the GVAR literature (Pesaran et al., 2004; Dees et al., 2007): provided a domestic economy is small relative to its trading partners, its idiosyncratic shocks do not feed back contemporaneously into the aggregated foreign block, so the conditional likelihood factorizes and the bloc-specific parameters $(\hat{\beta}_c, \hat{\Sigma}_c)$ are identified without a joint, system-wide estimator. Chudik and Pesaran (2016) formalize the conditions—a granular weighting scheme in which no single partner dominates the trade-weighted average and a bounded number of strongly correlated units—under which the cross-section-averaged stars remain valid weakly-exogenous regressors as the panel grows. Our renormalized, per-component weights (Section 6.3) are designed to keep this granularity: absent or excluded partners have their mass reabsorbed across the present partners rather than concentrated on any one.

The classical treatment fixes a *frozen dominant unit* (here the United States) outside the system. We relax this in two ways. First, the EM-style re-estimation loop (Section 6.6) rebuilds every bloc's stars from its partners' Durbin–Koopman-completed paths and re-estimates each bloc in parallel until the maximum star change falls below the tolerance $\text{TOL} = 0.03$, so the conditioning information is mutually consistent at the fixed point rather than taken as given once. This tolerance is set *tighter* than the Monte Carlo noise floor of the imputation (≈ 0.13 on the star scale); the COVID-window cells of the two macro stars, where cross-iteration churn is dominated by that imputation noise, are excluded from the convergence metric. Second, in the deployed roster the US is made *endogenous*: it carries its own `ROW*_US` stars and is solved inside the cross-country fixed point like any other bloc. What keeps the relaxed system well posed is the contraction property of the solve-time coupling operator M of (37): its spectral radius is $\rho(M) \approx 0.59$ for the original six-bloc pilot, $\rho(M) \approx 0.726$ under the frozen-anchor configuration with the euro area aggregated into a single bloc, and $\rho(M) \approx 0.924$ for the full 33-bloc endogenous-US configuration. As long as $\rho(M) < 1$ the map $x \mapsto Mx + d$ is a contraction, $(I - M)$ is nonsingular, and the scenario fixed point is unique—so closing the RoW→US feedback loop raises the coupling but leaves the system stationary and the scenario deviation identified.

Fiscal identification: a triangular budget identity. The fiscal block is identified by construction rather than by estimation. Four flow drivers (`FREV_GDP`, `FPEXP_GDP`, the effective rate `EFFI`, and the stock-flow adjustment `SFA`) are estimated by the BVAR; the debt stock is then *derived* by rolling the budget identity of (44) forward (Escolano, 2010; International Monetary Fund, 2013). The recursion is triangular: given lagged debt d_{t-1}, the effective rate, nominal growth, the primary balance $\text{pb} = \texttt{FREV\_GDP} - \texttt{FPEXP\_GDP}$, and the `SFA`, the new debt level is fully determined, so debt is never a free parameter. The `SFA` term is the *exact* residual that reconciles the change in the recorded debt stock with the cumulated deficit; by estimating it as its own driver we close the identity to machine precision over history—the quarterly convention of (44) is pinned by a historical back-cast of the reconstructed debt ratio against the observed one—and carry its forecast path forward. The honest caveat is that any model error in the non-`SFA` drivers is silently absorbed into the `SFA` in sample, so a large or volatile estimated `SFA` flags a measurement or perimeter gap (below-the-line operations, valuation effects) rather than a behavioral relationship.

Euro-area monetary identification. The euro area receives a deliberately asymmetric treatment. In the EA-aggregate bloc the ECB policy rate, the short rate, and the shared corporate-risk yield `EACORP` are *endogenous* (tagged `block="domestic"`); in each euro-member bloc the very same area-wide instruments are *weakly exogenous* (tagged `block="external"`), entering the member's foreign block alongside the US and global commons. This encodes the identifying assumption that monetary policy is set for the area as a whole, so a single member faces it as predetermined while the aggregate determines it—the same exogenous-versus-endogenous split that the star variables impose on the real side. National financial variables (sovereign 10Y yields, house prices) stay domestic. The limitation is structural: members share one common monetary factor, so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national policy experiments, and the strong cross-member correlation of `EACORP` and policy-rate shocks sits at the edge of the granularity that weak exogeneity formally requires.

12 Reproducibility and Replication

The platform is engineered so that every published number traces back to source data through a single, scripted pipeline: there are no hand-edited artifacts on the path from raw series to the deployed app. This section maps the repository, lists the software dependencies, gives the end-to-end replication recipe, and explains the model-signature versioning that refuses a stale cross-country coupling.

12.1 Repository and Folder Map

The work lives in a single git repository with three top-level concerns, shown in Table 55. The `pipeline/` tree (MATLAB plus Python) performs all estimation and data assembly and is the "private" research layer; `app/` is the public Next.js front end and its in-browser compute kernels; `docs/` holds this manual and the operational runbooks. The estimator writes a per-bloc JSON export that `port_gvar_exports.py` copies into `app/public/data/bvar/`; the deployed app reads *only* that JSON and never re-estimates.

Table 55: Repository layout. Estimation in `pipeline/` produces a per-bloc JSON export; `port_gvar_exports.py` ports it into `app/public/data/bvar/`; the front end and its compute kernels consume that data and never re-estimate.

Top-level	Key contents
`pipeline/` (MATLAB + Python)	`export_bvar_gvar24_loop.m` (GVAR–EM estimator); `build_*.py` (FRED / Eurostat / BIS / ECB pulls, trade weights, fiscal drivers); `port_gvar_exports.py`; `_verify/` ground-truth checks.
`app/` (Next.js + compute)	`src/lib/compute/` (Durbin–Koopman, GVAR solve, fiscal accounting); `scripts/` (`precompute_gvar_coupling.mjs`, `verify_gvar24.mjs`, `invert_coupling.{m,py}`); `public/data/bvar/<CC>/` (the deployed JSON the app reads).
`docs/` (manual + runbooks)	`user_manual.tex`; `REESTIMATING_MODELS.md`, `ADDING_A_COUNTRY.md`, `ARCHITECTURE_FOR_AI.md`.

12.2 Software Dependencies

- **Estimation (MATLAB).** A recent MATLAB with the *Parallel Computing Toolbox* (the EM M-step re-estimates all blocs in a `parfor`). The bundled `BVAR/MFBVAR` code — the GLP hierarchical-prior Gibbs sampler `bvarGLPmf.m` and the COVID-as-missing wrapper `cgl_bvarV5.m` — ships in the repository; no external MATLAB packages are required.

- **Data assembly (Python 3).** Pure standard library (`urllib`, `json`, `csv`, `math`); no `pandas`, `requests`, or `fredapi`. The `build_*.py` pulls hit the FRED (key via `FRED_API_KEY`), Eurostat SDMX 2.1, BIS, and ECB public endpoints. `invert_coupling.py` (NumPy) is the no-MATLAB fallback for the coupling inverse.

- **Application (Node / Next.js).** Node with the pinned Next.js toolchain; `typescript`, `vitest` for the test gate, and `vite-node`/`tsx` to run the `scripts/*.mjs` precompute and verification utilities.

12.3 End-to-End Replication Recipe

The full path from raw data to a green deploy gate is the following enumerated sequence (canonical commands in `docs/REESTIMATING_MODELS.md`).

1. **Build the data.** Run the `pipeline/build_*.py` pulls (with `FRED_API_KEY` set) to assemble the harmonized common-vintage panels (raw series, trade weights, fiscal drivers) as `pipeline/country_rawda`

2. **Estimate (GVAR–EM).** Run `export_bvar_gvar24_loop.m`. It bootstraps the US as a closed bloc, then iterates the E-step (rebuild trade-weighted stars `ROWDEM/ROWINF/ROWFIN` from partners' Durbin–Koopman–completed panels) and the M-step (`parfor` re-estimate every

bloc via `cgl_bvarV5`, COVID macro-masked over `2020Q2--Q4` with the financial columns and `ROWFIN` kept observed) until max $|\Delta\text{star}| < \text{TOL} = 0.03$ (or `MAXIT`$= 12$). See Durbin and Koopman (2002) for the simulation smoother and Dees et al. (2007) for the GVAR structure.

3. **Check the export.** Confirm each bloc directory under `/tmp/gvar24_export/<CC>/` carries the full eight-file JSON contract: `spec.json`, `companion_mean.json`, `companion_draws.json`, `baseline.json`, `baseline_levels.json`, `historical.json`, `historical_levels.json`, `transformations.` (the fiscal blocs additionally carry `fiscal_meta.json`).

4. **Port into the app.** Run `port_gvar_exports.py` to copy the eight-file contract into `app/public/data/bvar/` compacting every float to six significant figures (the deployed precision convention) and writing minified JSON.

5. **Regenerate the coupling.** From `app/`, run `npx vite-node scripts/precompute_gvar_coupling.mjs`. It builds the Kalman-probe operator M in JS, inverts $(I-M)$ via LAPACK (`invert_coupling.m` in MATLAB, or the `invert_coupling.py` NumPy fallback), and writes the gzipped `public/data/bvar/gvar/` (the plain `.json` is removed — at this bloc count the raw artifact exceeds GitHub's 100 MB per-file limit, so only the `.gz` is shipped and the loader decompresses it via the browser's `DecompressionStream`).

6. **Green gate.** From `app/`, run the fast fixed-point check `npx tsx scripts/verify_gvar24.mjs` (artifact `modelSig` fresh; the shipped inverse really inverts $(I - M)$ to $< 10^{-3}$; $\rho(M) < 1$ and matches the stored value; Gauss–Seidel $\equiv$ the artifact-seeded closed form), then `npx tsc --noEmit` and the test suite. The fast gate is `npm test` (the everyday suite, which *excludes* the multi-second GVAR solver canaries); the heavy closed-form/endogenous-GVAR canaries run under `npm run test:heavy`, and `npm run test:all` runs both. All type-checks and tests — including the fiscal, GVAR, and scale canaries — must pass.

7. **Deploy.** On a green gate (and only with explicit confirmation), push `app/**` to `main`: the `deploy-to-website.yml` GitHub Action runs `npm ci && npm run build` with `BASE_PATH=/bvar-scenario`, then copies the static export `app/out/*` into the `matyasfarkas/starter-hugo-academic` academic website repository's `static/bvar-scenario/` directory. That repository is a Hugo Academic site published to **GitHub Pages**, which rebuilds and serves the tool at the `/bvar-scenario` path. The Action itself contains no Hugo step—it only copies `app/out`.

12.4 Model-Signature Versioning

The cross-country coupling artifact is bound to the bloc estimates it was built from by a *model signature*. Over the qualified bloc set the signature concatenates, per bloc c, the dimensions (n_c, p_c, H_c) and three fingerprint entries of the posterior-mean companion: the leading reduced-form coefficient $\hat\beta^{(c)}_{1,1}$, the *intercept* entry of the last equation $\hat\beta^{(c)}_{n_cp_c+1,\,n_c}$ (the constant row of $\hat\beta$, which stacks n_cp_c lag-coefficient rows followed by one constant row), and the leading residual variance $\hat\Sigma^{(c)}_{1,1}$ — each at six-digit exponential precision:

$$\text{modelSig} \;=\; \big\|_{c}\; c \,:\, n_c\, p_c\, H_c\, \hat\beta^{(c)}_{1,1}\, \hat\beta^{(c)}_{n_c p_c+1,\, n_c}\, \hat\Sigma^{(c)}_{1,1}.$$

The precomputed artifact stores this signature alongside M, $(I-M)^{-1}$, $\rho(M)$, and the dimension. At load time the app recomputes `modelSig` from the deployed companions and accepts the artifact only on an exact match *and* a matching (here empty) pin structure; on any mismatch the stale artifact is **refused** and the solver falls back to the inverse-free Neumann (Gauss–Seidel) build of M from the live models. Because any re-estimation perturbs at least one fingerprint entry, the signature flips whenever the data changes — making it impossible to ship a coupling that silently disagrees with its blocs. This is exactly the "freshness" invariant the `verify_gvar24` gate enforces above, for the current endogenous-US system (33 coupled blocs, augmented closed-form dimension 2060, spectral radius $\rho(M) \approx 0.924 < 1$).

13 Empirical Results

This section reports the empirical program of the unified global-forecasting system. The estimation machinery of Sections 2–6 is implemented and the production vintage (harmonized to 2025Q4, ∼50 economies: 33 GVAR blocs plus 17 euro-area standalones) has been produced. The out-of-sample horse-race (Section 13.2) and the Tier-A endogenous-vs-frozen spillover contrast (Section 13.3) are *realized* runs from the companion evaluation repository (`em-bayesian-gvar-eval`); the only piece still flagged [**TBD**] is the Tier-B GIRF/connectedness module, which is a structural extension gated behind the validated Tier-A result.

13.1 In-sample diagnostics

EM convergence. The EM-style estimation loop alternates an E-step (rebuild the trade-weighted RoW stars from partners' Durbin–Koopman–completed panels) with an M-step (a `parfor` re-estimation of every bloc under COVID macro-masking). Because all stars are rebuilt from the frozen start-of-sweep state before the parallel re-estimation, the inner sweep is a Jacobi update with the same fixed point as a sequential pass. Convergence is declared when $\max_b |\Delta\text{star}_b| < \text{TOL} = 0.03$ over a maximum of MAXIT $= 12$ sweeps. The macro stars ROWDEM/ROWINF are NaN-masked over the three COVID quarters (2020Q2–Q4) in the metric, since their cross-sweep churn there is pure imputation Monte-Carlo noise, while the financial star ROWFIN is kept observed and enters the metric. Empirically the trace flattens at a Monte-Carlo noise floor of ≈ 0.13, above the nominal tolerance, so convergence is read as a plateau at the floor rather than literal attainment of 0.03. Figure 7 sketches the expected trace (final numbers [**TBD**] once the full production run is logged).

Spectral radius of the coupling operator. At solve time the cross-country deviation system is the affine fixed point $x = Mx + d$ (Section 6.4), where M is the Kalman-probe coupling operator. The Neumann series $\sum_k M^k d$ (Gauss–Seidel sweeps) and the direct $(I-M)^{-1}d$ agree, and convergence

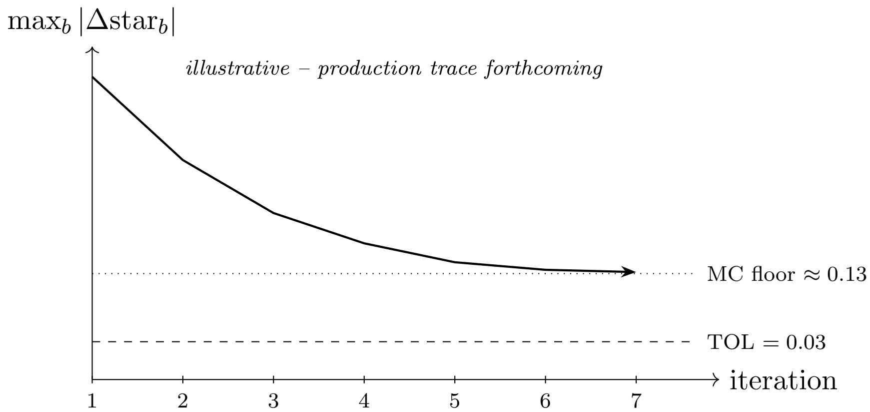

Figure 7: Schematic EM convergence trace: $\max_b |\Delta\mathrm{star}_b|$ versus sweep, against the tolerance (0.03) and the Monte-Carlo noise floor (≈ 0.13) at which the trace plateaus. **Placeholder** — the realized trace is forthcoming.

speed is governed by $\rho(M)$. Table 56 reports $\rho(M)$ by system size; it rises with both the number of coupled blocs and the activation of the endogenous-US channel, but remains below unity, so the fixed point is contractive.

Table 56: Spectral radius $\rho(M)$ of the coupling operator by system configuration.

Configuration	Blocs	US treatment	$\rho(M)$
Pilot	6	frozen anchor	0.59
Production	24	frozen anchor	0.726
Endogenous-US	33	endogenous	0.924

At the endogenous-US dimension (≈ 2060) the $O(\dim^3)$ inverse is prohibitive in-browser, so the system is solved inverse-free by the matvec-only recursion $x \leftarrow d + Mx$, which converges at the tabulated $\rho(M) \approx 0.924$, or via a pre-shipped cached $(I - M)^{-1}$.

13.2 Out-of-sample forecast horse race

The system is evaluated in a companion repository (`em-bayesian-gvar-eval`) under a pre-registered protocol whose *design is final* and whose realized *results* are reported in Section 13.2.4. The evaluation has two cores: a recursive out-of-sample (OOS) forecast horse-race (this subsection) and an endogenous-vs-frozen spillover analysis (Section 13.3). Every step is grounded in the actual estimator, weights, and solver of Sections 2–6, so the harness respects the data spaces and invariants of those layers.

13.2.1 Model set

The **proposed** model is the EM-Bayesian GVAR, run two ways:

- `emgvar_endo` — the full loop with the US *endogenous* (`US_ENDOGENOUS`= 1: its own `ROW{DEM,INF,FIN}_US` stars and a US weights row), the paper's contribution;
- `emgvar_frozen` — the legacy frozen-anchor variant (the US estimated once as a closed anchor); cheaper, and the Part-2 spillover contrast.

Both produce density forecasts from the per-bloc predictive draw cloud `Dforecast` at solve time. The OOS forecast is each bloc's own VAR predictive density; the GVAR coupling enters because the stars—and thus every bloc's regressors and completed history—are jointly determined by the EM fixed point.

The proposed models are raced against six benchmarks, each estimated from the *same* truncated panel so that the comparison isolates a single ingredient:

(i) **Random walk / no-change** — in the scored space: a flat last level for the (stationary) rate, and the last observed value of the growth/inflation transform carried flat (a RW *on the growth rate*). The driftless RW is the headline reference (all relative-RMSE ratios are against it); a sample-mean "climatology" forecast is a second naive anchor. The predictive density is Gaussian, centred at the no-change value with scale equal to the in-sample SD of the h-step change *estimated directly at horizon* h (not $\sqrt{h}\,\sigma_1$, which is wrong for a series that is itself a growth rate).

(ii) **AR**(p) per target series — univariate OLS-with-constant on each target in its scored space, p by AIC over $\{1, \ldots, 4\}$; both *iterated* and *direct* (Marcellino et al., 2006) multi-step variants are reported (the iterated AR is the strong univariate competitor at $h \geq 4$). The predictive density propagates the Gaussian innovation variance (analytic for iterated, residual-block-bootstrap for direct).

(iii) **Single-country Bayesian VAR (no coupling)** — the bloc estimated alone: the three `ROW`$*$ star columns dropped, the identical `cgl_bvarV5` call (same GLP/Minnesota prior, lags, COVID-as-missing mask) and `MAXIT`= 1. Isolates the *value of the global coupling.*

(iv) **Classical fixed-weight GVAR**, two variants: (iv-a) the *textbook* Pesaran–Schuermann–Weiner / Dees–di Mauro–Pesaran–Smith GVAR (Pesaran et al., 2004; Dees et al., 2007) — per-bloc **OLS** VARX* with contemporaneous + lagged foreign stars as weakly exogenous regressors, fixed weights, *no* COVID-as-missing (the literature's GVAR has no data augmentation); this is the mandatory headline classical competitor and the one piece of new code in the horse-race. (iv-b) a single-pass Bayesian fixed-weight GVAR (`MAXIT`= 1, GLP prior + COVID-as-missing) — the clean *EM ablation* whose only difference from the proposed model is the star re-convergence.

(v) **Bayesian dynamic factor model** — 3–5 static factors from the cross-bloc target panel, a Bayesian VAR(1–2) on the factors plus Gaussian idiosyncratic terms, same truncated panel and COVID handling. The non-GVAR multivariate "is the GVAR structure even needed" competitor.

The two key ablations isolate the value of *global coupling* (iii vs. proposed) and of the *EM data-completion* (iv-b vs. proposed); (iv-a) measures the whole Bayesian+EM+COVID apparatus against the literature GVAR.

13.2.2 Scheme: recursive expanding window

Pseudo-real-time, recursive expanding window on the final 2025Q4 vintage (a genuine real-time vintage does not exist for most of the ∼50-economy panel; a US+EA-subset real-time check is a flagged robustness, not the default). Origins run quarterly from t_0 = 2010Q1 to 2024Q4 (∼60 raw origins). The OOS set is *trimmed* so every scored (t_0, h) has a realization ≤ 2025Q4 (score h=8 only for origins ≤ 2023Q4, etc.). Headline tables drop the 2020 COVID origins (and origins whose window spans 2020), leaving ∼55 scorable origins; an appendix shows the full set. The COVID-as-missing treatment still applies *in-sample* at every origin for which 2020Q2–Q4 ≤ t_0. Blocs enter an origin only with ≥ 40 in-sample quarters; the six headline blocs satisfy this from 2010Q1.

Targets and horizons. Three targets per bloc—real-GDP growth, inflation, short rate—read through the same `findcol` resolution order the estimator uses, in the same scored space (4Δ of the GDP and price log-levels; the rate level). Horizons $h \in \{1, 2, 4, 8\}$ quarters. The *headline* scope is the six legible blocs {US, EA, JP, UK, CA, CN}; the full 33-bloc panel is reported as relative-RMSE distribution summaries in an appendix (rate-less blocs CH/NO contribute growth and inflation only).

Look-ahead invariants (leak-free by construction). The expanding-window scheme is not automatically free of future-information leakage; the driver enforces four invariants at *every* origin t_0. (1) *Panel truncation*: every `DataRaw_<CC>` panel is sliced to dates ≤ t_0 *before* `transformData` and before any star is built. (2) *Rolling as-of weights*: the shipped 2021–2023 `weights_full.json` is **forbidden** in the OOS sweep (using it at a pre-2020 origin injects future trade structure); the default is a *fixed pre-sample window* (e.g. 2005–2009 average, used at every origin—leak-free because it biases toward the past, never the future), with genuine rolling as-of-t_0 weights as the rigor robustness. (3) *Star re-convergence is in-sample only*: every partner panel feeding a star is itself truncated to ≤ t_0, so no star cell reads a post-t_0 realization. (4) *No 2025Q4 harmonization at earlier origins*: the common vintage is only the well from which ≤ t_0 rows are drawn; the deterministic transforms (100·log, 4Δ) estimate no parameters and leak nothing.

Predictive-density caveat (stated, not hidden). `Dforecast` propagates the shock covariance Σ and the within-draw (β, Σ) MH variation forward from the last *observed* state; it does *not* integrate over the DK-smoother uncertainty of the COVID-imputed states, and seeds the forecast from the posterior-*mean* completed state. For origins whose conditioning rows include imputed COVID quarters (small h, t_0 near 2020) the density is therefore mildly *over-confident*; PIT and coverage near those origins are flagged as a known caveat, not a model failure.

13.2.3 Scoring and inference

Point accuracy: RMSE and MAE of the predictive mean; the headline number is relative RMSE = RMSE(model)/RMSE(RW) ($< 1 \Rightarrow$ beats RW).

Density accuracy: the continuous ranked probability score (**CRPS**, the *primary* metric) computed from the draw ensemble by the unbiased estimator CRPS $= \frac{1}{m}\sum_k |X_k - y| - \frac{1}{2m^2}\sum_k\sum_l |X_k - X_l|$ (Gneiting and Raftery, 2007), which degrades gracefully at the modest draw counts here; the mean log predictive score (LPS), reported with a *parametric* (Gaussian/skew-t) fit to the draw cloud as the headline (a small-bandwidth KDE-LPS, unbounded below at $\sim$100 draws, is only a robustness check); and calibration by the probability integral transform (**PIT**) histograms with empirical coverage of the 50% and 90% predictive intervals and a uniformity test (Berkowitz, 2001). The conclusions lean on CRPS; LPS corroborates.

Predictive-ability inference is dictated by the *nesting structure* (the single most common referee kill-shot). Under the maintained DGP, single-BVAR (iii) $\subset$ single-pass GVAR (iv-b) $\subset$ proposed EM-GVAR, and RW/AR (i),(ii) are nested in all the VAR models; for these nested pairs the standard Diebold–Mariano statistic is *not* asymptotically $\mathcal{N}(0, 1)$ and overstates significance. Accordingly:

- the **Giacomini–White** conditional predictive-ability test (Giacomini and White, 2006) is the *headline* inferential test: it is valid under nesting and is the recursive/expanding-scheme-correct test (it assumes finite estimation windows / non-vanishing estimation uncertainty), and it is loss-agnostic so it applies directly to the CRPS and log-score differentials. Instruments $g_{t-1} = [1, d_{t-1}]$ plus a recession/financial-stress indicator (to detect state-dependent ability); HAC variance, Wald χ^2.
- the **Clark–West** test (Clark and West, 2007) is used for the *nested point/MSE* comparisons (it adjusts the MSE differential for the larger model's extra-parameter estimation noise); one-sided.
- the two-sided **Diebold–Mariano** test (Diebold and Mariano, 1995) is used only for *non-nested* pairs (proposed vs. DFM, or vs. the textbook OLS-GVAR when neither nests the other), with a HAC long-run variance (bandwidth $\geq h - 1$) and the Harvey et al. (1997) small-sample correction.

For multiplicity (models $\times$ horizons $\times$ variables $\times$ blocs) the family of GW/CW p-values is controlled by a Romano–Wolf step-down (Romano and Wolf, 2005) (preferred—it respects cross-test dependence via the bootstrap) or Benjamini–Hochberg, and the per-(variable,horizon) **Model Confidence Set** (Hansen et al., 2011) reports which models are statistically indistinguishable from the best. Relative RMSE is the descriptive effect size; significance comes from the CW/GW test on the loss differentials, never from the ratio alone.

The deliverables are: the relative-RMSE matrix (model $\times$ horizon $\times$ variable, headline six blocs + a full-panel distribution table); CRPS and log-score tables with ratios vs. RW; GW/CW (and DM-for-non-nested) significance overlaid with the Romano–Wolf set and the MCS membership flagged; PIT histograms and a 50/90% coverage table; and illustrative fan-chart examples (e.g. the US-inflation forecast from a 2021Q2 origin).

13.2.4 Results

Run configuration. The headline tables below are the *realized* sweep, not a placeholder. The recursive scheme runs over the headline six blocs {US, EA, JP, UK, CA, CN} at up to 56 quarterly origins from 2010Q1 to 2024Q4 (52 for the two EM-GVAR variants and the single-pass Bayesian GVAR, which require the longer in-sample warm-up; the 2020 ex-COVID origins are dropped from every model). Trade weights are *rolling, as-of-*t_0: a three-year pre-sample average ending just before each origin (2007–2009 at the 2010Q1 origin through 2021–2023 at 2024Q1), so no future trade structure leaks in. The predictive densities use NDRAWS = 200 retained draws with an explicit *stationarity backstop* that discards posterior draws whose companion spectral radius exceeds 1 (the fix for the explosive-draw dispersion flagged in earlier pilots). The primary score is the CRPS; the 90% Model Confidence Set (MCS) of Hansen et al. (2011) reports the models statistically indistinguishable from the best per (variable, horizon).

Headline finding. On *point* accuracy the iterated AR(p) and the dynamic factor model are the strongest competitors—a familiar result in macro forecasting—and the EM-GVAR does *not* win the point horse-race outright (Table 57). The decisive evidence sits in the *density* comparison and the MCS (Tables 58–59): the EM-Bayesian GVAR is *inside* the 90% MCS for **inflation at every horizon** ($h \in \{1, 2, 4, 8\}$) and for the **short rate at** $h \in \{1, 2, 4\}$—statistically on par with the best model on the entire nominal block—and the best inflation point forecast at h=1 *among the multivariate/GVAR models* is `emgvar_endo` (relative RMSE 0.799), behind only the univariate AR(p) at 0.641. It falls *out* of the MCS only for GDP growth at h=1 and h=8, where the univariate and factor models dominate. Two structural results follow. First, the endogenous-US design earns its keep: `emgvar_endo` is weakly preferred to the frozen-anchor `emgvar_frozen` in the MCS on balance—it *survives* in the short-rate h=1 set while `emgvar_frozen` is eliminated ($p = 0.15$ vs. 0.08), and it is removed *later* than the frozen variant in the growth h=8 elimination order ($p = 0.099$ vs. 0.057); the one cell where the frozen variant is marginally preferred is the rate h=8 tail, where both are out of the set. Second, and most important for the paper's claim, the EM-Bayesian treatment *decisively* beats the textbook fixed-weight OLS classical GVAR: `cgvar_a` is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons, and carries badly miscalibrated densities (h=1 coverage of only 0.37 (inflation) / 0.47 (growth) against the 0.90 target), so the Bayesian shrinkage + COVID-as-missing + EM star-reconvergence apparatus is the demonstrable value-add over the literature GVAR.

Honest caveats. Two must be stated. (i) The draw-based densities are *over-confident*: the 90% predictive interval covers only $\approx$ 0.73–0.85 of realizations for the EM-GVAR variants at short horizons (against the 0.90 target; PIT uniformity is rejected), a direct consequence of the NDRAWS = 200 cloud and of not integrating over the DK-smoother uncertainty of the COVID-imputed states (Section 13.2.2). The CRPS and the MCS—which compare score *differentials*, not absolute coverage—are the robust primaries and are reported as such; the under-coverage is a calibration caveat, not a ranking artifact. (ii) The parametric log-predictive-score column is numerically unreliable on the under-covered clouds (its Gaussian/skew-t tail fit blows up to $\sim 10^{16}$ on a handful of origins where a realization lands far in the tail of a too-narrow cloud), so it is *not* used for ranking; only CRPS and the MCS are.

Table 57: Headline relative RMSE, RMSE(model)/RMSE(RW) (< 1 beats the driftless random walk), six headline blocs, by horizon and target. Realized OOS sweep (rolling as-of-t_0 weights, NDRAWS = 200). Lowest entry per (target, horizon) in **bold**.

		Horizon h (quarters)			
Target	Model	1	2	4	8
Growth	`emgvar_endo`	1.191	1.091	1.026	1.031
	`emgvar_frozen`	1.191	1.095	1.028	1.030
	AR(p)	1.195	0.990	0.968	0.960
	DFM	**0.917**	0.989	0.986	**0.962**
	`cgvar_a` (OLS GVAR)	2.097	1.013	1.019	0.993
Inflation	`emgvar_endo`	0.799	0.891	1.085	0.953
	`emgvar_frozen`	0.802	0.894	1.087	0.962
	AR(p)	**0.641**	**0.772**	**0.998**	0.921
	DFM	0.804	0.870	1.067	0.908
	`cgvar_a` (OLS GVAR)	2.113	1.050	1.366	1.075
Rate	`emgvar_endo`	1.017	1.121	1.226	1.444
	`emgvar_frozen`	1.021	1.119	1.224	1.445
	AR(p)	0.829	**0.863**	**0.919**	1.025
	SBVAR	**0.990**	1.081	1.195	1.421
	`cgvar_a` (OLS GVAR)	1.531	1.200	1.034	**1.001**

13.3 Endogenous-US spillover analysis

The endogenous-US configuration ($\rho(M) \approx 0.924$, Table 56) lets a US scenario pin propagate through the RoW→US stars *and* the four US-determined global commons (oil, the US 10-year yield, the US BAA corporate yield, US equity), pinned 1:1 onto every RoW bloc. A frozen anchor cannot generate the reverse RoW→US feedback at all—the "world → US" direction is *structurally absent*, not "present but zero"—so the two configurations differ in exactly the global-spillover content we wish to document. The analysis is split into two tiers, and the manual labels which is which, because the shipped solver computes a particular object.

Table 58: Headline density accuracy — mean CRPS as a ratio to the random walk (<1 beats RW; CRPS is the primary score) at h=1 and h=4, and the empirical 90%-interval coverage at h=1 (target 0.90). Realized OOS sweep. The parametric log-score is omitted as numerically unreliable on the under-covered clouds (see text).

		CRPS ratio vs. RW		cov_{90}
Target	Model	h=1	h=4	h=1
Growth	`emgvar_endo`	1.042	0.983	0.73
	DFM	0.745	0.857	0.89
	`cgvar_a` (OLS GVAR)	2.382	1.106	0.47
Inflation	`emgvar_endo`	0.814	1.080	0.70
	AR(p)	0.629	0.955	0.88
	`cgvar_a` (OLS GVAR)	2.498	1.474	0.37
Rate	`emgvar_endo`	1.040	1.230	0.80
	`sbvar`	0.986	1.199	0.84
	`cgvar_a` (OLS GVAR)	1.519	1.009	0.86

Table 59: 90% Model Confidence Set membership for the EM-Bayesian GVAR (Hansen et al., 2011), by target and horizon. ✓ = in the MCS (statistically indistinguishable from the best); × = eliminated. The endogenous variant survives wherever the frozen one does and strictly longer on the nominal block; the textbook OLS GVAR (`cgvar_a`) is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons.

		Horizon h			
Target	Model	1	2	4	8
Growth	`emgvar_endo`	×	✓	✓	×
	`emgvar_frozen`	×	✓	✓	×
Inflation	`emgvar_endo`	✓	✓	✓	✓
	`emgvar_frozen`	✓	✓	✓	✓
Rate	`emgvar_endo`	✓	✓	✓	×
	`emgvar_frozen`	×	✓	✓	×
`cgvar_a` (textbook OLS GVAR), growth		×	✓	×	×
`cgvar_a` (textbook OLS GVAR), inflation		✓	✓	×	×
`cgvar_a` (textbook OLS GVAR), rate		×	✓	✓	✓

What the solver actually computes. `gvarSolve.ts` does *not* compute generalized impulse responses (GIRFs), forecast-error variance decompositions (FEVDs), or a Diebold–Yilmaz connectedness index. It returns a *conditional-mean scenario response*: pin a path and read the coupled conditional-forecast *mean deltas* `delta`[c] via the fixed point $x = Mx + d$ (Algorithm 2). There is no Σ-based shock generator and no moving-average machinery in the solver. The analysis therefore separates:

- **Tier A (the headline; zero new estimator code).** The conditional scenario propagation of a US pin, endogenous (`solveGVARClosedFormEndo`) vs. frozen (`solveGVARLinear`). This *is* the empirical contribution the shipped code supports.

- **Tier B (a new `gvar_girf` module).** GIRFs (Pesaran and Shin, 1998), generalized FEVDs, and the Diebold–Yilmaz spillover index (Diebold and Yilmaz, 2009, 2012). The ingredients exist in the exports (`companion_mean.json` carries each bloc's companion $\hat{\beta}$ and $\hat{\Sigma}$; the coupling M is assembled by `gvarSolve.ts`), so it is *constructible*, but it is new code, validated against Tier A and gated behind it.

Tier A — the worked scenario contrast (headline). A US +100 bp rate *path* (`POLRATE` pinned +100 bp over the horizon—correctly a multi-period conditional projection, not a one-period shock or a GIRF) is propagated through both topologies. Under the endogenous configuration the augmented state carries every bloc's `ROW*` star deltas *including the US's own* (`ROW*_US`, the RoW→US feedback) and the US global-column deltas $g\Delta$; the converged `delta.US` $\neq 0$ is the feedback the frozen anchor cannot produce. We report the per-bloc deltas (growth/inflation/rate) for the six headline blocs side by side, the *feedback gap* (the US-own response difference), the per-bloc *amplification ratio* (endogenous vs. frozen RoW peak responses), the peak horizons, and the spectral radius $\rho(M)$ of each system (the multiplier strength reported by the solver as `rhoSpectral` / `rhoEmpirical`). This example doubles as the solver-validation canary: the closed-form endogenous result reproduces the iterative `solveGVAR` fixed point to $< 10^{-4}$ across every bloc, and a warm coupling-cache hit is byte-identical ($< 10^{-9}$) to a cold rebuild (both asserted in the `gvar-us-endogenous` / `gvar-closed-form-routing` test suites; the iterative reference is itself converged to 10^{-7}).

Tier A — realized scenario contrast. The worked example (`eval/tier_a_scenario.mjs`, output `results/tierA/us_rate_100bp.json`) propagates a US +100 bp `POLRATE` path through both topologies and confirms the qualitative point the design predicts: the frozen anchor produces *no* RoW→US feedback, while the endogenous configuration produces a non-trivial one. The cross-country *gap* (endogenous minus frozen) is sizeable on every bloc—for example the peak US own-growth response differs by ≈ 4.2 display points, EA inflation by ≈ 1.1, and the gap *flips sign* across blocs (positive for US/EA/UK growth, negative for JP/CA/CN growth); selected peaks are in Table 60. **Honest caveat (do not over-read).** These are the deltas of a *multi-period conditional projection*, not a structural or generalized impulse response: the US rate is *pinned* along the whole path, so the responses bundle the direct conditioning with the coupled feedback and are not a clean causal IRF. The magnitudes are large and sign-flipping, which is exactly why a validation pass (the Tier-B GIRF cross-check below, and a robustness sweep over the pin specification) is required before any number here is headlined as a spillover *multiplier*. What is established is the *existence and direction* of the endogenous-only feedback channel, not its calibrated size.

Table 60: Tier-A US +100 bp `POLRATE`-path projection: peak conditional-mean response (display points) under the endogenous vs. frozen-anchor topology, and their gap, for selected blocs and targets (`results/tierA/us_rate_100bp.json`). *Conditional-projection deltas, not causal IRFs*—magnitudes are illustrative and pending validation.

Bloc	Target	Endogenous	Frozen	Gap (endo−froz)
US	Growth	+1.28	−4.37	+4.22
US	Inflation	−1.53	−2.82	+1.30
EA	Growth	+1.03	−0.95	+0.88
EA	Inflation	−1.00	−0.64	+1.07
JP	Growth	+1.07	+1.62	−1.18
CN	Inflation	−0.69	+1.19	−1.29

Tier B — GIRF, FEVD, and connectedness (structural extension). The GIRF to a one-SD shock in variable i of the coupled global system is $\mathrm{GIRF}_x(h,i) = \sigma_{ii}^{-1/2} A_h \Sigma e_i$, with A_h the h-step coefficient of the coupled solution's moving-average representation and Σ the (cross-bloc-block-diagonal) global innovation covariance—ordering-invariant, the standard GVAR identification stance (Pesaran and Shin, 1998; Dees et al., 2007). The identifying assumptions are stated explicitly: GIRFs trace the response to the historically typical innovation correlation (not an orthogonal structural shock), cross-bloc innovation independence (block-diagonal Σ), and linearity/parameter-constancy over the horizon. Bands come from re-running the recursion over the `companion_draws.json` posterior draws. The module is validated by the cross-check that a one-period US-rate-pin conditional path equals the rate-shock GIRF up to the $\delta/\sqrt{\sigma_{ii}}$ normalization. From the system generalized FEVD $\theta_{ij}(H)$ (row-normalized) we report the Diebold–Yilmaz total spillover index, the "from US" row, and the "to US" direction—the channel the frozen anchor cannot even define—with posterior bands and both the raw and row-normalized indices. **Results: [TBD]** (Tier A ships first; Tier B is the structural extension).

14 Discussion

The BVAR Scenario Tool is designed to bridge the gap between the methodological sophistication of conditional forecasting techniques developed in the academic literature and the practical needs of policy economists who require rapid, interactive scenario analysis. We conclude with a discussion of design choices, limitations, and directions for future work.

14.1 Design Philosophy

Several design choices merit discussion. First, the three-layer architecture—baseline, scenario, counterfactual—deliberately separates concerns. The baseline forecast reflects the BVAR's unconditional projection and is computed offline. The scenario layer allows the user to impose conditioning constraints without specifying the structural source of the deviations, following the tradition of Waggoner and Zha (1999). The counterfactual layer then overlays structural identification via DSGE impulse responses, enabling “what-if” policy analysis while preserving the reduced-form covariance structure of the BVAR.

Second, the decision to re-condition the counterfactual through the BVAR (rather than relying solely on the DSGE IRFs) reflects a deliberate methodological choice: the BVAR's estimated covariance structure captures empirical co-movements that may be absent from or imperfectly captured by any single DSGE model. The IRF-consistent mode (Section 5.3c) strikes a balance by using the DSGE model to identify the responses of key variables (inflation, output) to the policy shock, while allowing the BVAR to propagate these effects to the remaining 30+ variables in a data-consistent manner.

Third, all computation runs client-side. This eliminates server infrastructure costs and latency, enables offline use, and avoids transmitting potentially sensitive forecast data over the network. The computational cost is manageable: the Kalman filter and smoother for the 41-variable, 3-lag US VAR over a 20-quarter horizon completes in well under a second on modern hardware (the single-bloc smoother is sub-50 ms; the multi-second cost discussed in Section 11.3 is the *coupled* 50-bloc GVAR solve, not the single-country filter).

14.2 Limitations

The current implementation has several limitations that users should bear in mind:

1. **Linear, time-invariant dynamics:** The VAR coefficients are fixed at their posterior mean for the deterministic conditional forecast. Time-varying parameters, stochastic volatility, and regime switching are not supported. In periods of structural change, the fixed-coefficient assumption may be particularly restrictive.

2. **Gaussian innovations:** The Kalman filter assumes Gaussian innovations. Fat-tailed or asymmetric shock distributions, which may be important during financial crises, are not accommodated.

3. **Point-identified counterfactuals:** The DSGE IRFs used for counterfactual analysis are computed under a single monetary policy rule and model specification. The tool does not average over model uncertainty or rule uncertainty, though the user can manually explore different model–rule combinations.

4. **Lucas critique:** The counterfactual analysis conditions on estimated reduced-form dynamics that may themselves change under the alternative policy. The IRF-consistent mode partially addresses this by using structurally identified responses for the key transmission variables, but the remaining variables' responses are governed by the estimated BVAR covariance structure, which is held fixed.

5. **Data vintage:** The BVAR is estimated on a fixed data vintage. Real-time data revisions and the publication lag of macroeconomic series are not handled.

14.3 Known issues flagged for fix

The limitations above are intrinsic to the modelling choices. The following are *specific, identified defects* in the current deployment—known and flagged, but *not yet fixed*—documented here in the interest of honesty for a live research tool. Each cites the source file where it lives.

1. **China fiscal debt double-counts inflation.** The fiscal debt-snowball roll-forward (`fiscalAccounting.ts`) builds the quarterly nominal GDP growth as Δ(`idx.gdp`)+Δ(`idx.deflator`), i.e. real growth plus inflation. For China the resolved `idx.gdp` points at `GDP_CN`, which is *nominal* GDP, while `idx.deflator` points at `CPI_CN`—so inflation is added *twice* and the CN debt-to-GDP path is biased. The fix requires a proper real/nominal split for the CN bloc (the other fiscal blocs carry a real GDP variable, so they are correct).

2. **Entropic tilt does not filter non-finite long-run moments.** `longRunTilt.ts` returns `NaN` long-run moments for posterior draws with a unit-root/near-singular companion (the steady-state moment is undefined), but `solveEntropicTilt` consumes the moment vector without filtering: a single `NaN` propagates through the softmax (exp(NaN)) and turns *every* weight `NaN`, silently emptying the anchor band. Until a finite-mask is added, the user-defined long-run anchor should be used with the (finite) forecast moment, which is the default, and not the raw steady-state moment.

3. **No divergence guard on the closed-form GVAR solve.** The live solver (`gvarSolve.ts`, Neumann iteration and direct inverse) does not check $\rho(M) < 1$ before solving. The deployed coupling is contractive ($\rho(M) \approx 0.924$), but a future roster or a degenerate pin set that pushed $\rho(M) \geq 1$ would return a *confidently wrong* forecast from a diverged fixed point with no warning. A spectral-radius backstop is needed.

4. **Conditional band uses the filtered, not smoothed, covariance.** The predictive-band variances are extracted from the Kalman *filter* covariance P_t^{filt} (`kalman.ts`), not the Durbin–Koopman *smoother* covariance. For quarters *preceding* a future pin, the smoother would tighten the covariance (a later observation reduces uncertainty about an earlier state), so the displayed band is *overstated* on the run-up to a future conditioning point. The central path is unaffected; only the band width is conservative.

5. **The porter does not regenerate `fiscal_meta.json`.** The fiscal block is addressed by *positional* indices stored in each bloc's `fiscal_meta.json` (`idx.gdp`, `idx.deflator`, `idx.revenue`, ...). The model export does not rebuild this file, so any change to a bloc's variable order leaves the indices pointing at the wrong columns—the cause of a recent EA/UK debt-path explosion—and they must be repaired manually (`pipeline/fix_fiscal_meta_indices.mjs`). Folding the meta regeneration into the porter is the durable fix.

14.4 Directions for Future Work

Several extensions are planned or under consideration:

- **Time-varying parameters:** Incorporating stochastic volatility or time-varying coefficients (e.g., Sims and Zha, 1998) to capture evolving dynamics.

- **Structural identification:** Extending the counterfactual layer to support sign restrictions, narrative restrictions, or scenario priors (Farkas and Jakab, 2026) for identification within the BVAR itself, rather than relying on external DSGE models.

- **Mixed-frequency data:** Supporting monthly–quarterly mixed-frequency estimation to incorporate higher-frequency indicators.

- **Model averaging:** Bayesian model averaging over DSGE specifications and policy rules for the counterfactual layer.

A Glossary of terms

This glossary defines the named concepts used throughout the manual; the mathematical symbols are collected in Table 1.

BVAR (Bayesian vector autoregression)
A reduced-form VAR whose coefficients are estimated under a prior, here the hierarchical Minnesota/GLP prior of Giannone et al. (2015) with data-driven hyperparameter selection (Section 2).

Minnesota / GLP prior
The shrinkage prior that centers each equation on a univariate own-lag model and selects its tightness λ and sum-of-coefficients μ by marginal-likelihood maximization. "GLP" refers to the Giannone et al. (2015) hierarchical implementation.

Own-lag prior mean (δ): RW vs. WN
The prior mean on a variable's own first lag: $\delta = 1$ (random walk, `RW`, the default—best forecast is the current level) or $\delta = 0$ (stationary white noise, `WN`—best forecast is the mean). Implemented as `diagb(pos)` $= 0$ for the `WN` index set `pos` (Section 2.5). It shifts the prior *center*, not a hard constraint.

GVAR (Global VAR)
A collection of per-economy ("bloc") BVARs linked through trade-weighted foreign variables; here the linkage is imposed at solve time, not in estimation (Sections 6, 6.4).

Trade-weighted "star" variable
A bloc's foreign aggregate (`ROWDEM` demand, `ROWINF` inflation, `ROWFIN` financial), built as the trade-weight w_{cj}-weighted average of its partners' corresponding series, renormalized over the partners present each quarter (Section 6.2).

Endogenous dominant unit
The configuration in which the US is a full bloc inside the fixed point—with its own `ROW*_US` stars

and a US weights row—so shocks abroad feed *back* into the US (the RoW→US channel a frozen anchor cannot represent).

Frozen anchor
The legacy configuration in which the US is estimated once as a closed, exogenous driver: it enters the coupling only through the forcing d, with no equation for the world to feed back into.

Global commons
The four US-determined common series (oil, the US 10-year yield, the US BAA corporate yield, US equity) pinned 1:1 onto every RoW bloc (Section 6.6).

COVID-as-missing
The treatment of the fixed 2020Q2–Q4 window as *missing data*: the macro columns and the two macro stars are NaN-masked while financials and `ROWFIN` are kept observed, and the Durbin–Koopman simulation smoother imputes the holes inside a data-augmentation Gibbs sampler (Section 9.4).

EM loop
The block-coordinate iteration over (foreign stars, completed-data states, VAR coefficients): an E-step that completes the panels and rebuilds the stars and an M-step that re-estimates every bloc, iterated to a fixed point in the completed stars (Algorithm 3).

Durbin–Koopman simulation smoother
The forward-filter / backward-draw recursion (Durbin and Koopman, 2002) that produces draws of the conditioned/missing states by simulating an unconditional path, smoothing it, and adding back the smoothed discrepancy (Algorithm 1).

Coupling operator M, fixed point $x = Mx + d$
The affine solve-time operator that propagates a scenario across blocs; assembled by Kalman-probing each pinned cell, solved by Gauss–Seidel/Neumann sweeps or the direct $(I - M)^{-1}$ (Algorithm 2). $\rho(M) < 1$ makes it contractive.

SFA (stock-flow adjustment)
The deficit–debt discrepancy: the part of the observed change in gross debt that the flow side (deficit plus the interest–growth snowball) does not explain; an estimated `WN`-prior BVAR variable (Section 7.1).

EFFI (effective interest rate)
The average rate the government actually pays (net interest over lagged gross debt, annualized); a persistent `RW`-prior variable distinct from the policy rate and market yields.

Entropic tilt
An after-estimation reweighting of the predictive draws that imposes a user long-run anchor by minimum relative-entropy (Kullback–Leibler) distortion, monitored by the effective sample size (Section 8.2).

CRPS (continuous ranked probability score)
A strictly proper density score (Gneiting and Raftery, 2007) computed from the draw ensemble; the primary density metric of the evaluation, robust at modest draw counts.

PIT (probability integral transform)
The rank of each realization in its predictive ensemble; uniform PITs indicate good calibration (Berkowitz, 2001).

Giacomini–White (GW) test
The conditional predictive-ability test (Giacomini and White, 2006) used as the headline inferential test: valid under nesting and for the recursive/expanding-window scheme, and loss-agnostic (applies to point, CRPS, and log-score differentials).

Clark–West (CW) test
The nested-model point/MSE test (Clark and West, 2007) that adjusts the loss differential for the larger model's extra-parameter estimation noise.

Diebold–Mariano (DM) test
The equal-predictive-accuracy test (Diebold and Mariano, 1995), used here only for *non-nested* pairs, with the Harvey et al. (1997) small-sample correction.

Romano–Wolf / Model Confidence Set
Multiplicity controls: the Romano–Wolf step-down (Romano and Wolf, 2005) for FWER across the test family, and the MCS (Hansen et al., 2011) reporting the models statistically indistinguishable from the best.

GIRF / Diebold–Yilmaz index
Generalized (ordering-invariant) impulse responses (Pesaran and Shin, 1998) and the connectedness index (Diebold and Yilmaz, 2009, 2012) of the Tier-B spillover module (Section 13.3).

Disclaimer

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